General Chemistry I

Chemistry: The central science

Matter and Measurement

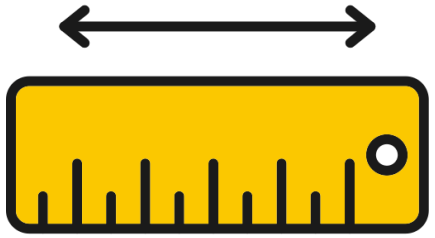
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nguyen-duc.anh@usth.edu.vn





Contents

* Classification of matter
* Physical and chemical changes, separation of mixture
* Unite of measurement

Objectives

* Use the basic vocabulary of matter and energy

* Distinguish Chemical *versus* Physical properties;

Chemical *versus* Physical changes

* Recognize various forms of matter

* Understand the concepts: mixture, substance, compound, molecule, element, and atoms

Matter

Matter is anything that has mass and occupies space.

Mass is a measure of the quantity of matter

in a sample of any material

Weight (≠ Mass) of an object is the force that

is exerted on its mass by gravity





Astronaut

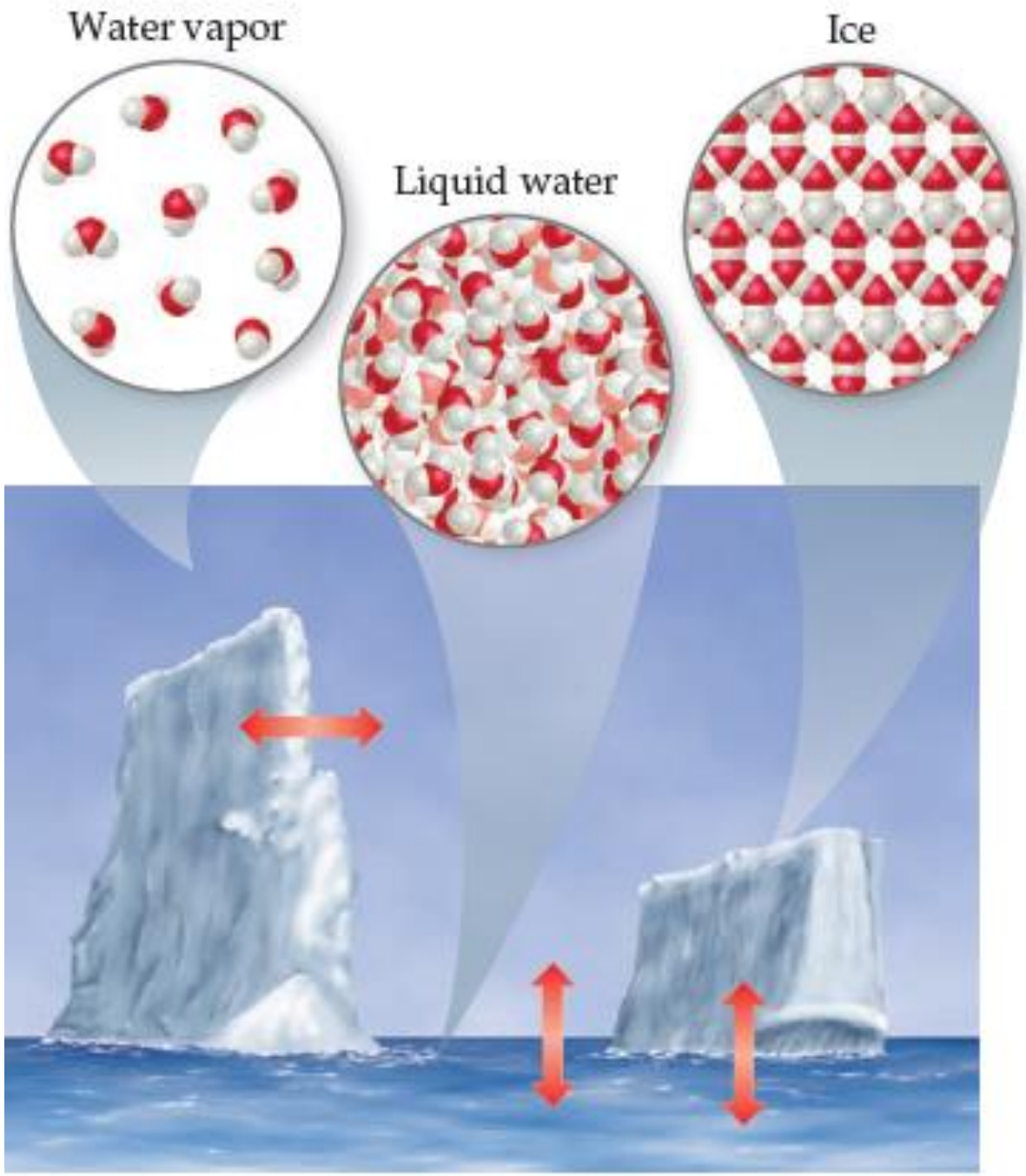
Mass: identical on the Planet and in Space

Weight ≈ 0 in Space

Classification of Matter

Physical states: Gas, Liquid, Solid

Chemical composition: Element, Compound, Mixture


Water vapor
Liquid water
Ice

Gas (Vapor)



Air, like all gases, takes the shape of its container and expands to fill its container.





Gas (Vapor): No fixed of volume or shape

uniformly fills its container

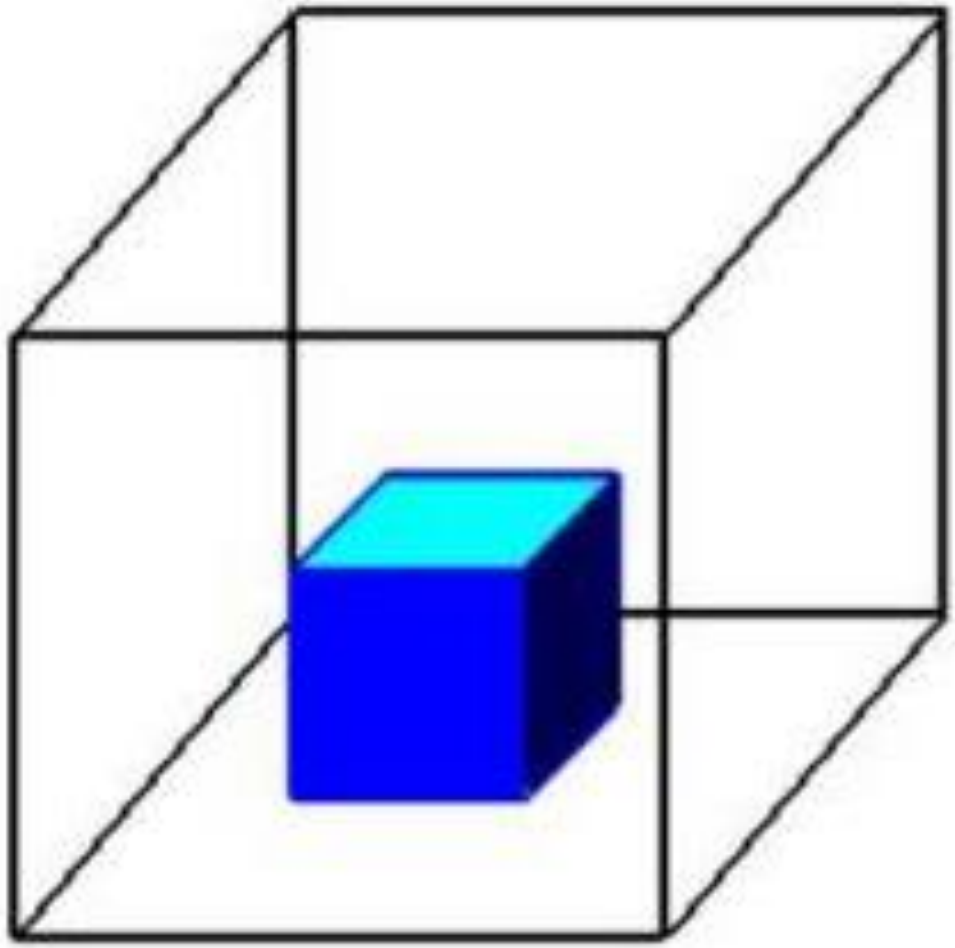
Liquid:

* Has distinct volume independently of its container

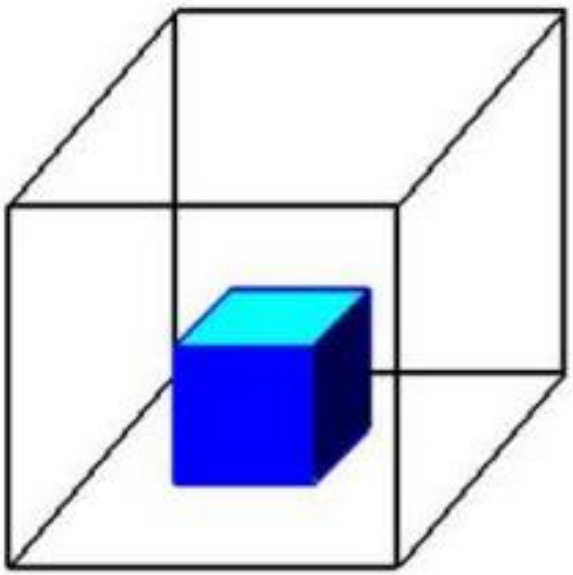
* Assumes a shape of a portion of its container



Solid: definite shape and volume



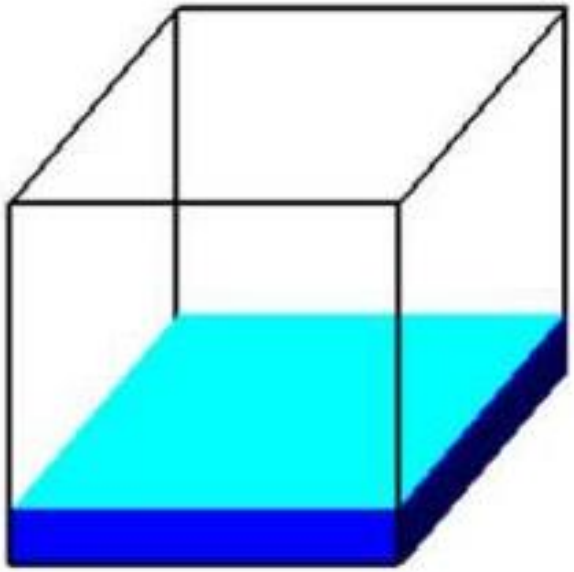
Physical States (Phases) of Matter



Solid

Holds Shape

Fixed Volume

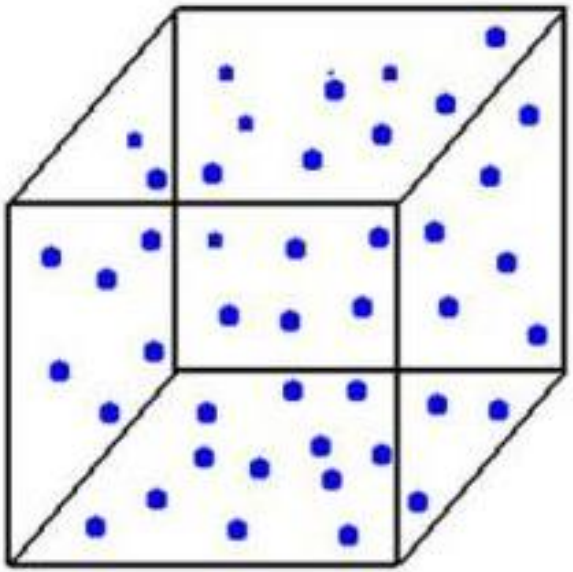


Liquid

Shape of Container

Free Surface

Fixed Volume



Gas

Shape of Container

Volume of Container



Iodine



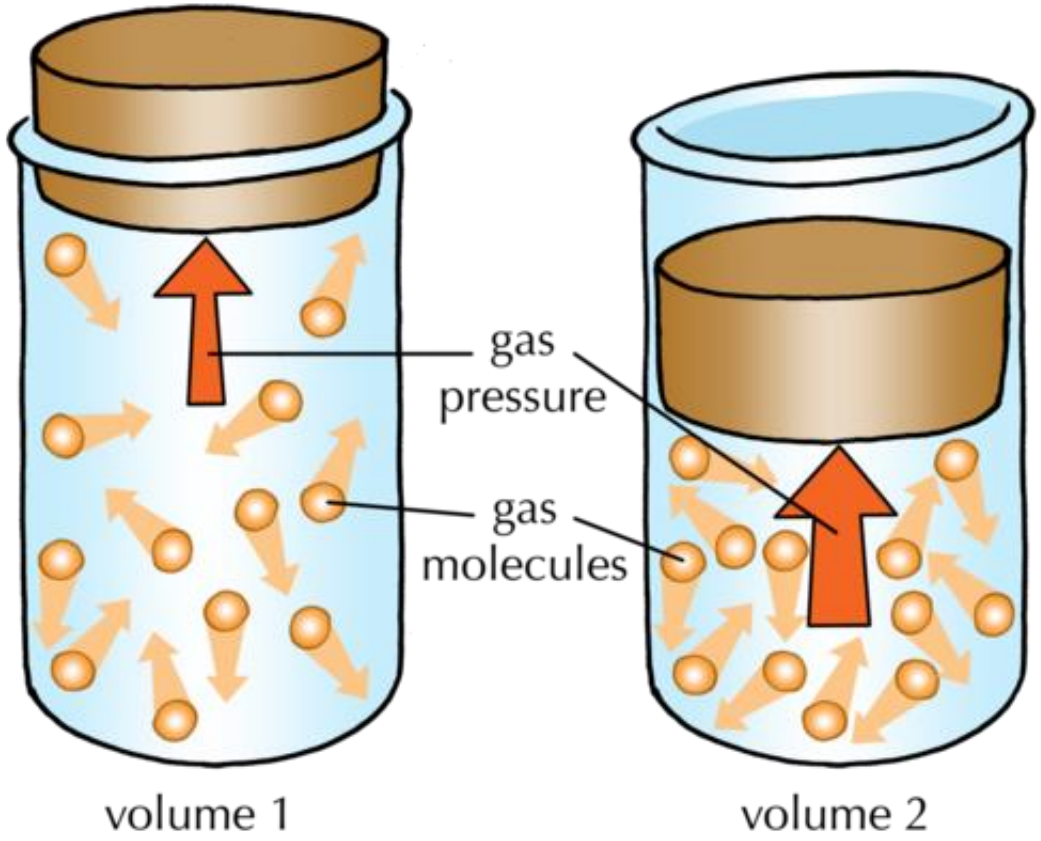
Bromine



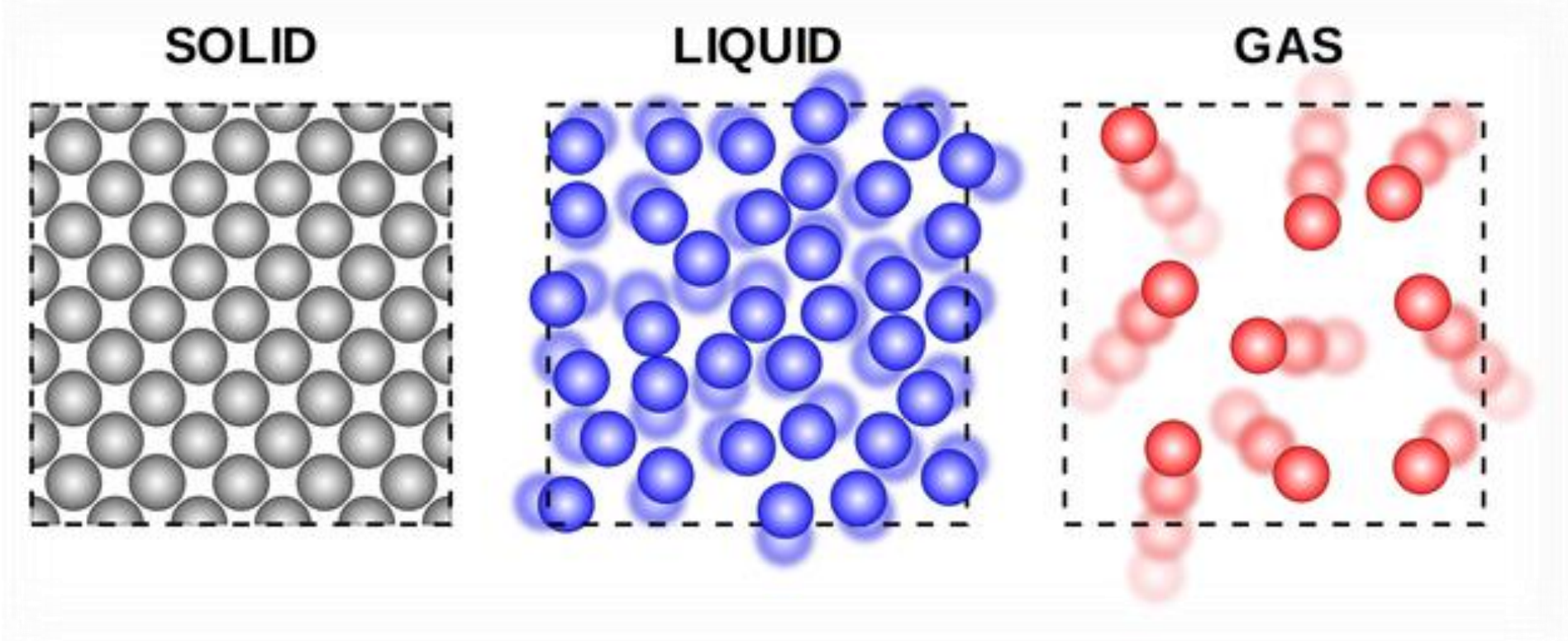
Chlorine

Gas can be compressed to smaller volume/ expanded to large volume

(Not possible with liquid and solid)



Physical States: Look at molecular level

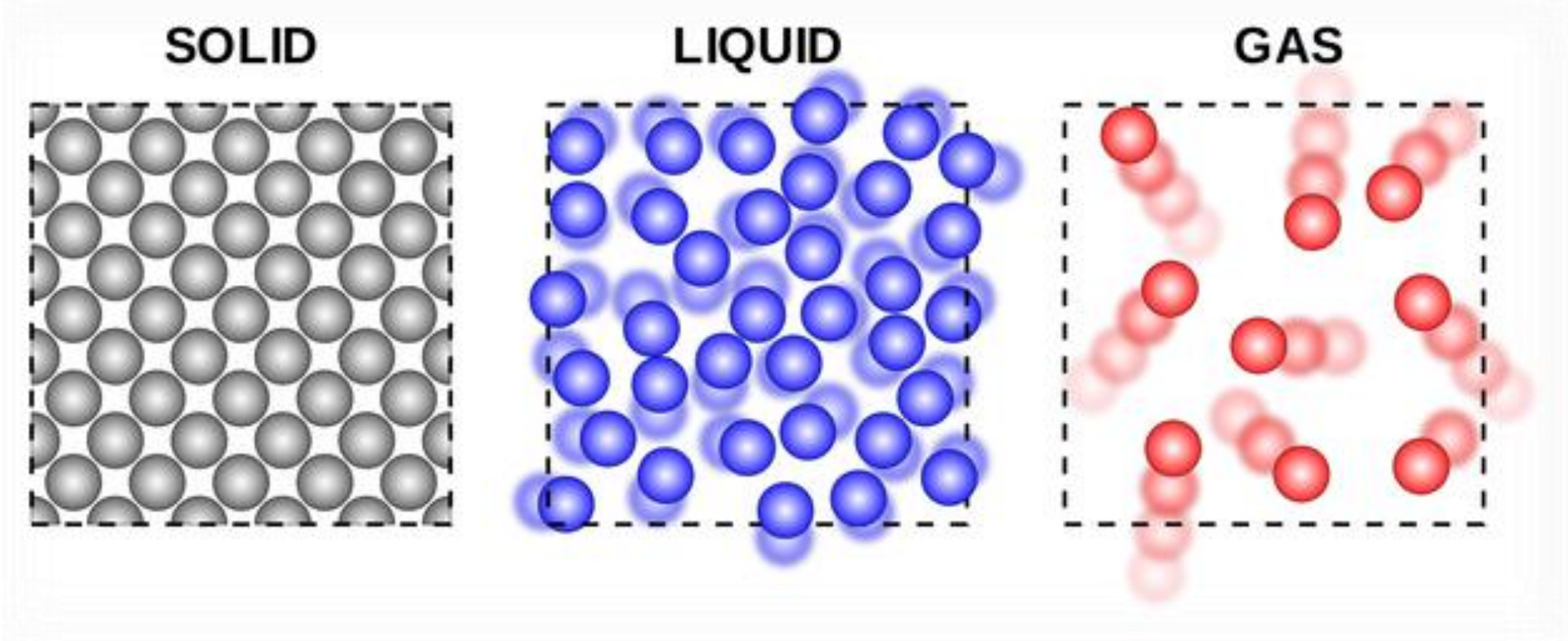


+ Position of molecules

+ Movement/ Interaction of molecules

+ Distance between molecules

Phase transformation

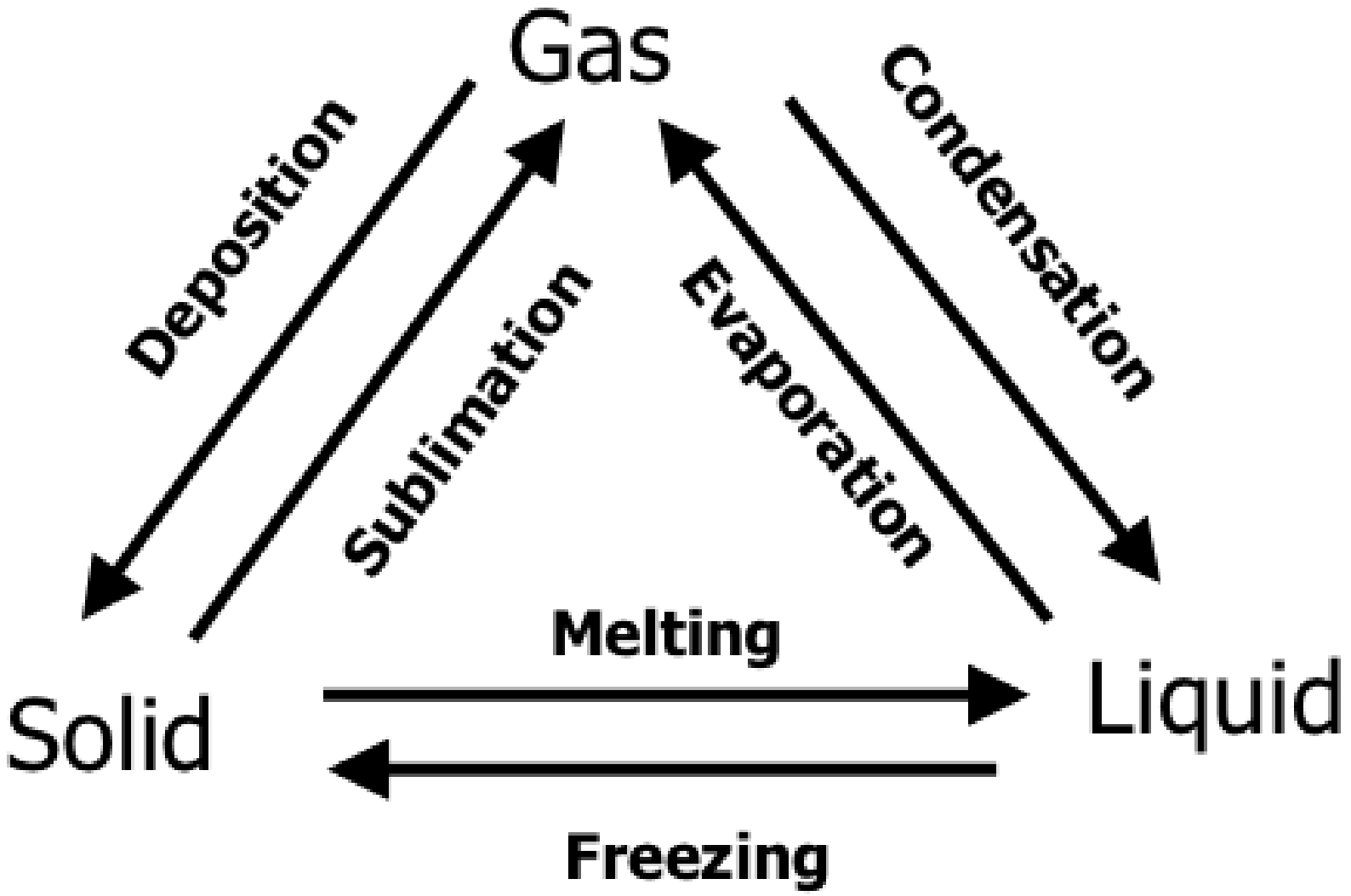


+ **Temperature**: Changes in movement of molecules

+ **Pressure**: Changes in molecular distances/ interaction

Experiment: *Cooling an air-balloon by liquid nitrogen*

Phase transformation



Welcome to Paris…

A typical winter day…Snowing….Accidents









How can we “remove” snow (*especially on the street/ highway*)?



How can we melt the ice/ snow?

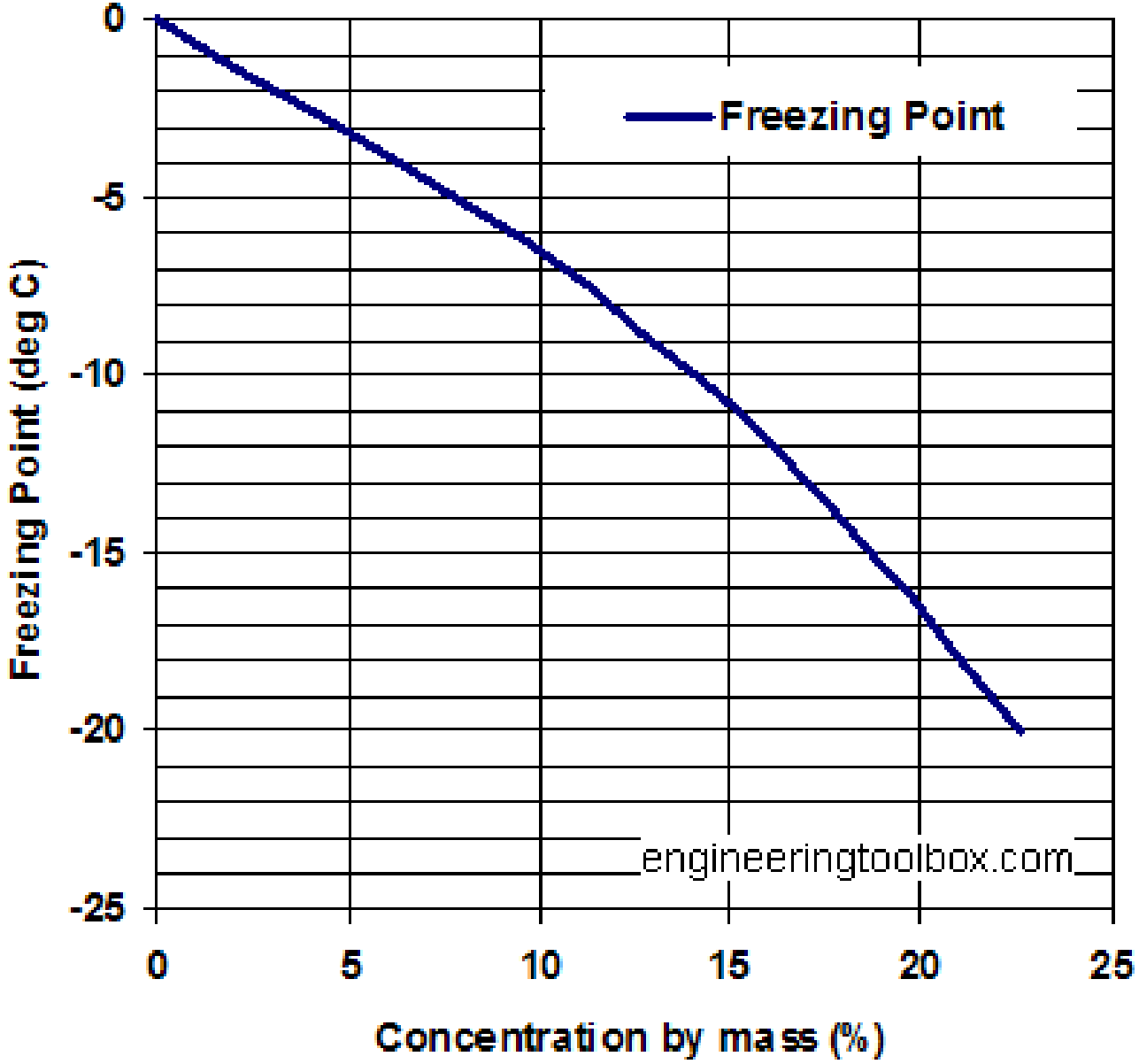


…Putting Salt (NaCl)



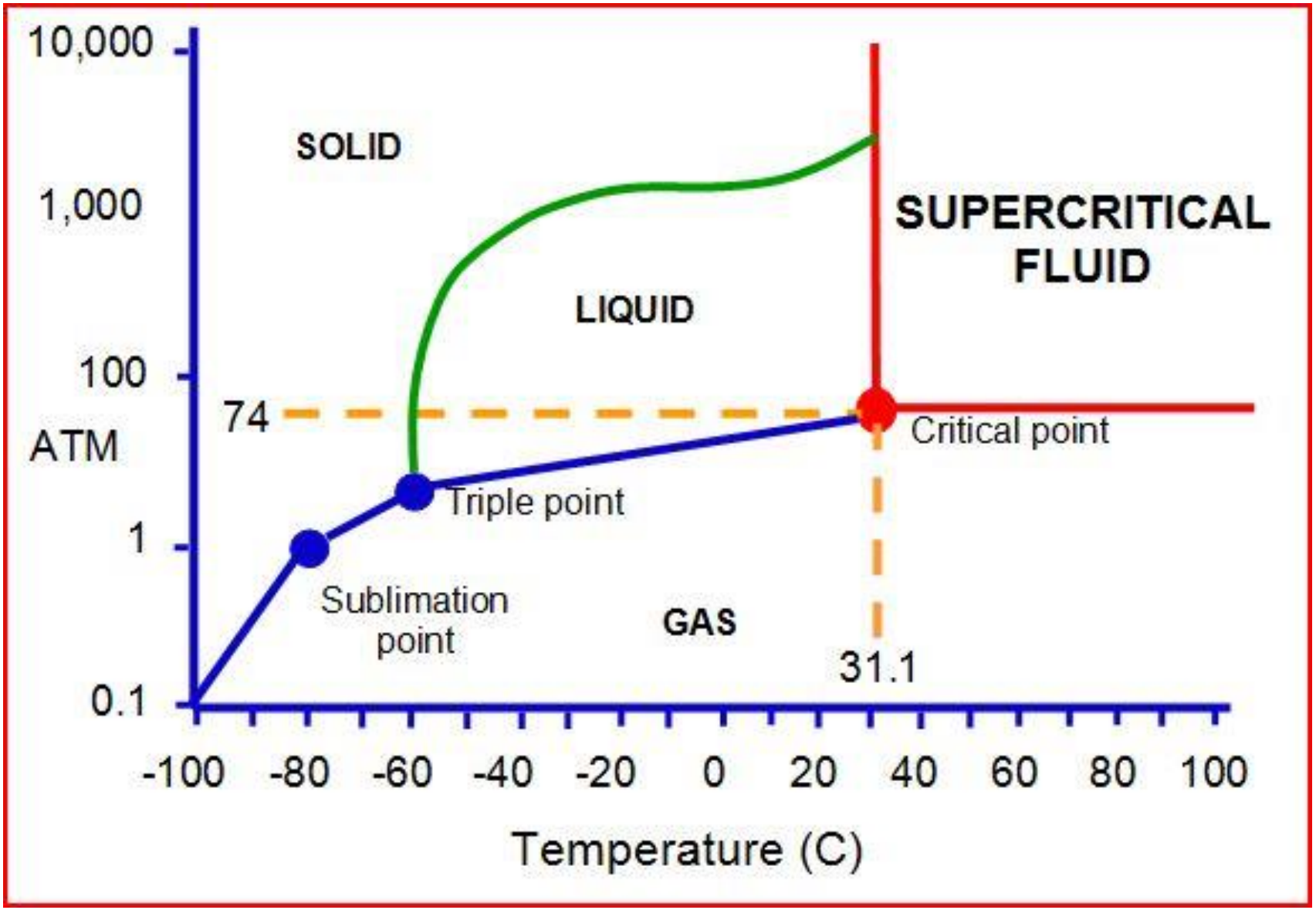


Freezing point of a NaCl solution



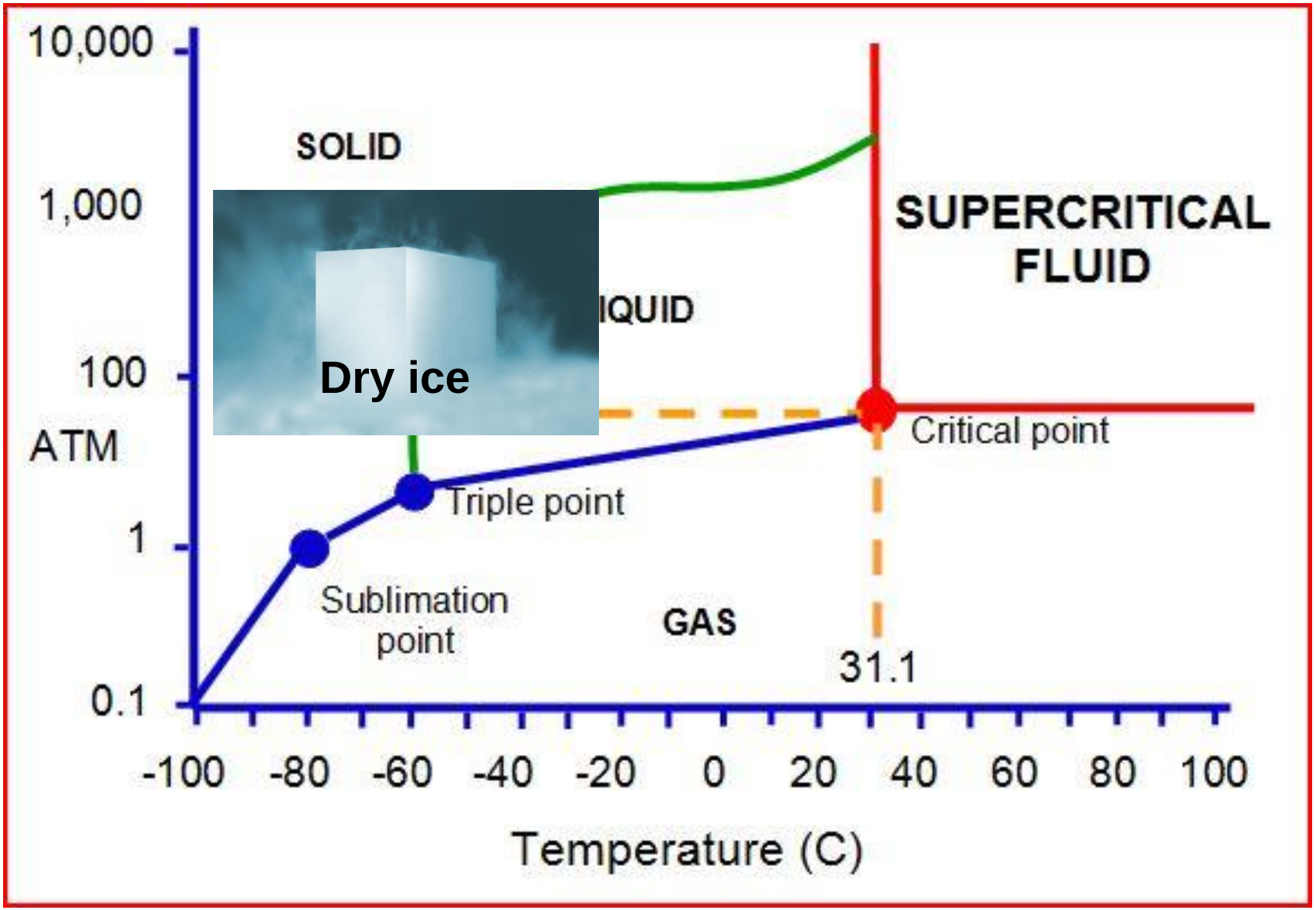
Supercritical CO_2 (sCO_2)

CO_2 phase diagram

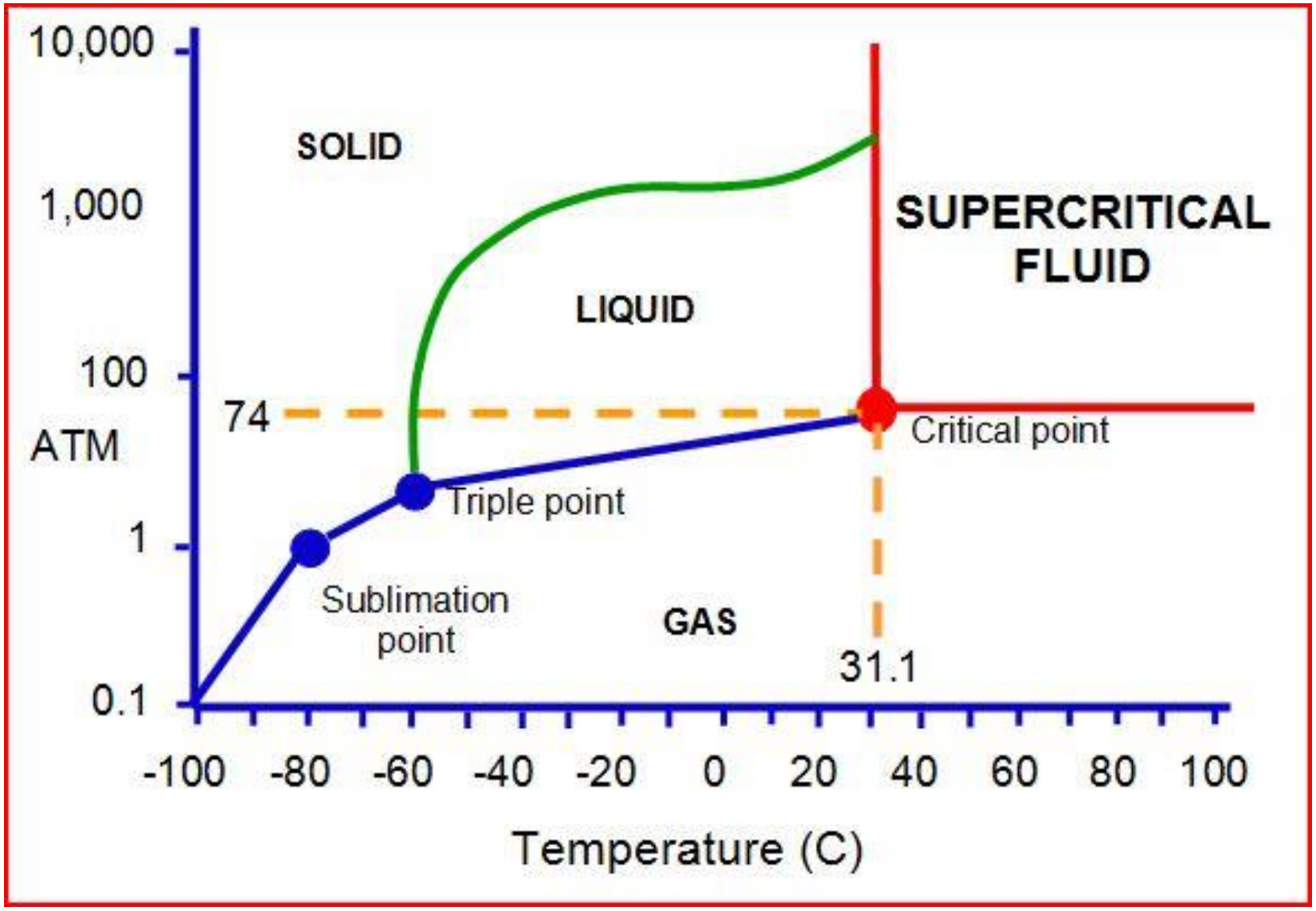


Supercritical CO_2

CO_2 phase diagram

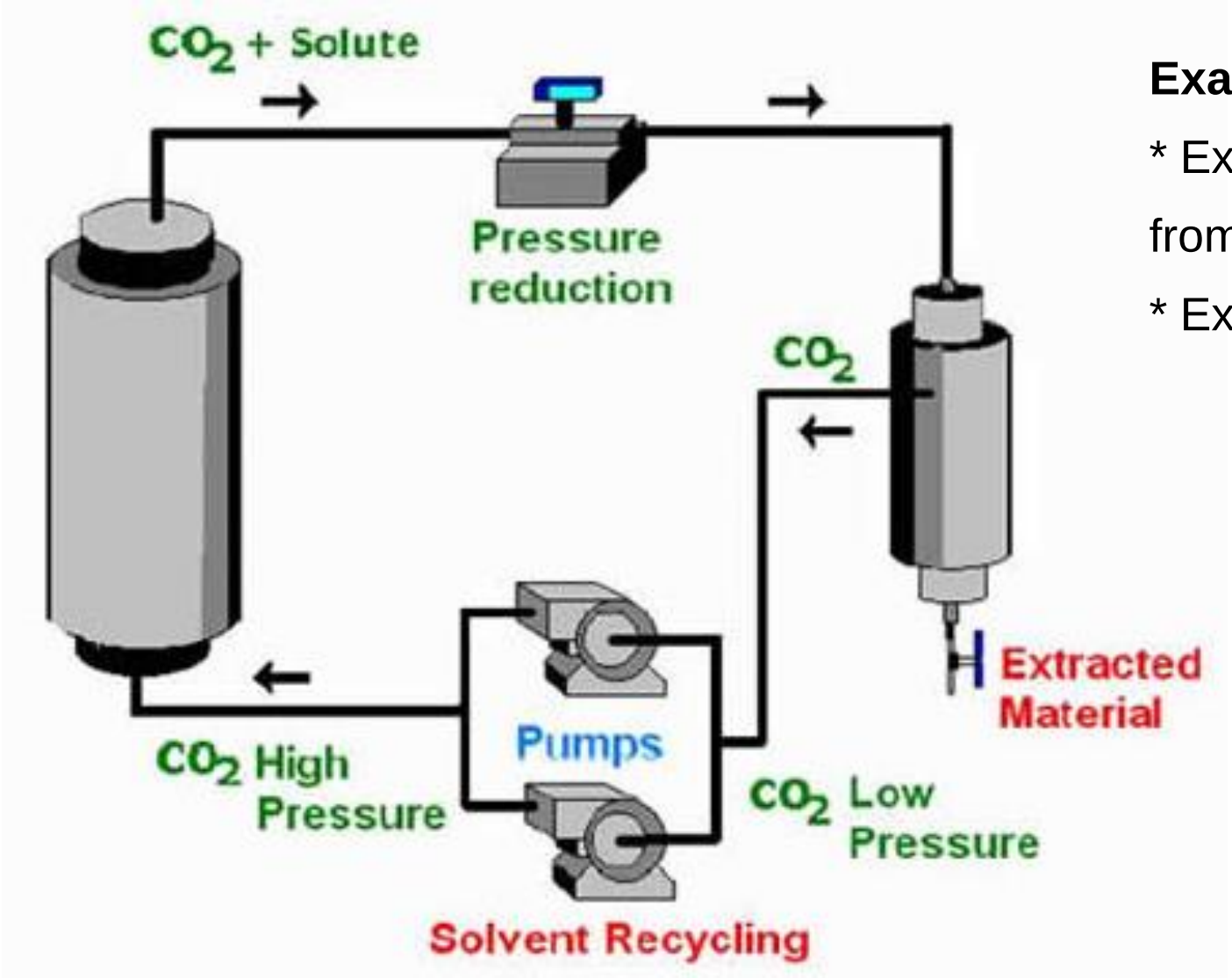


Supercritical CO_2: distinct liquid and gas phases do not exist. It can diffuse through solids like a gas, and dissolve materials like a liquid



Application: Green Extraction of Natural Products

(Use of organic solvents is no longer needed!)



Examples:

* Extraction of caffeine from coffee and tea
* Extraction of fish oil

Natural product extraction by using supercritical CO_2

https://www.youtube.com/watch?v=AHhJkahRkPw

Self-learning….

Extraction principles: “Like dissolves Like”

Polar solvent dissolves polar compound

Non-polar solvent dissolves non-polar compound

+ What is polarity of a **Supercritical CO_2 (**sCO_2)?

e.g. in comparison with ethanol?

+ Can we tune the polarity of a supercritical CO_2?

Classification of Matter

Physical states: Gas, Liquid, Solid, Plasma

Chemical composition: Element, Compound, Mixture

Definition

Pure Substance (*in short*: Substance)

Matter that has distinct composition and properties that does not vary from sample to sample (*or* Lab to lab)



Well, please make sure that you got a pure substance

Mixture:

A combination of **two or more substances** in which each substance retains its chemical identity

NO chemical reaction occurs

Examples:

Air (homogeneous mixture): $O_2 + N_2 + CO_2 + H_2O$

Ethanol/ water mixture (homogeneous mixture) **(Solution)**

Oil/ water mixture (heterogeneous mixture)

Homogeneous mixture *versus* **Heterogeneous mixture?**

Oil/ water mixture (heterogeneous mixture)



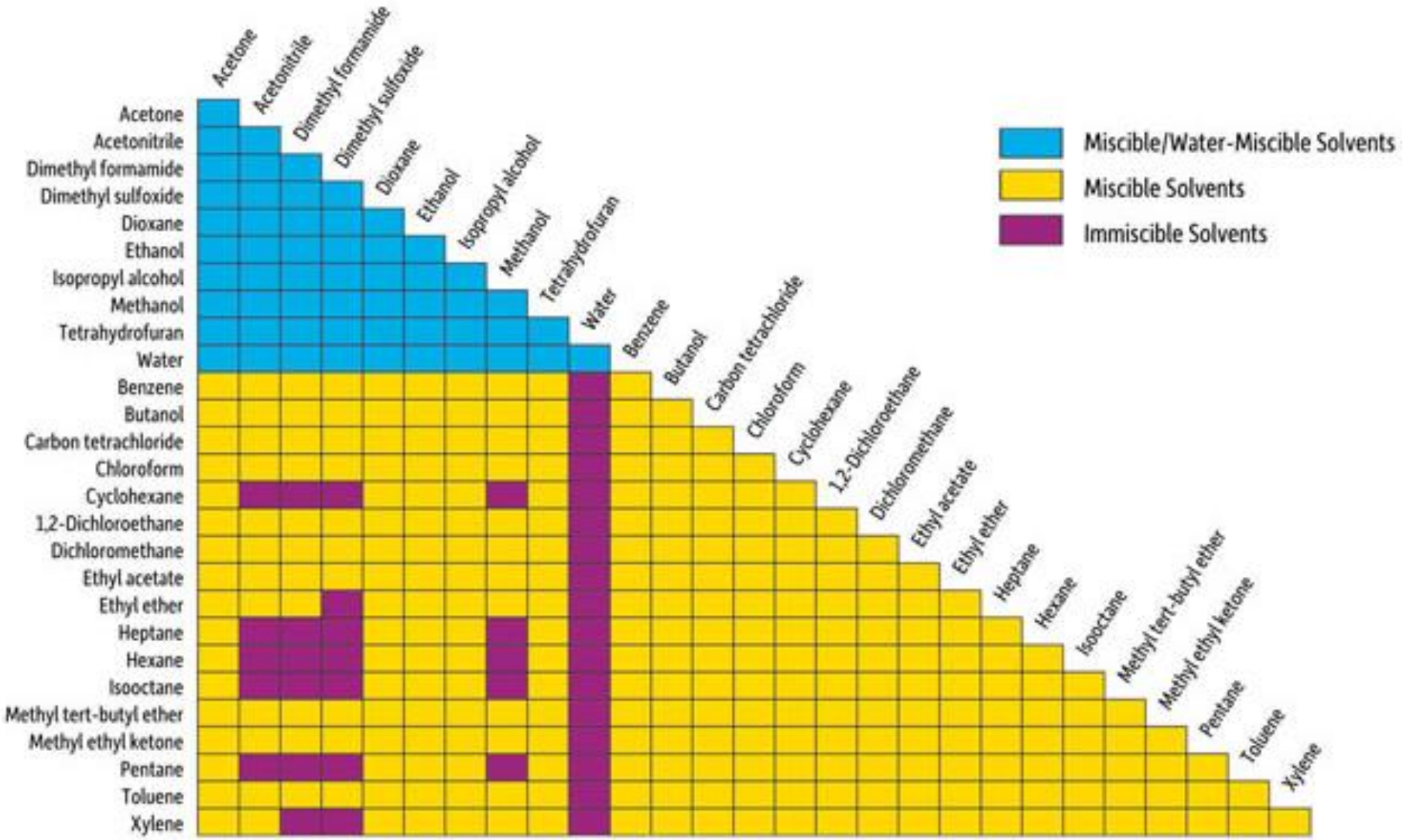
How do they wash their hands?



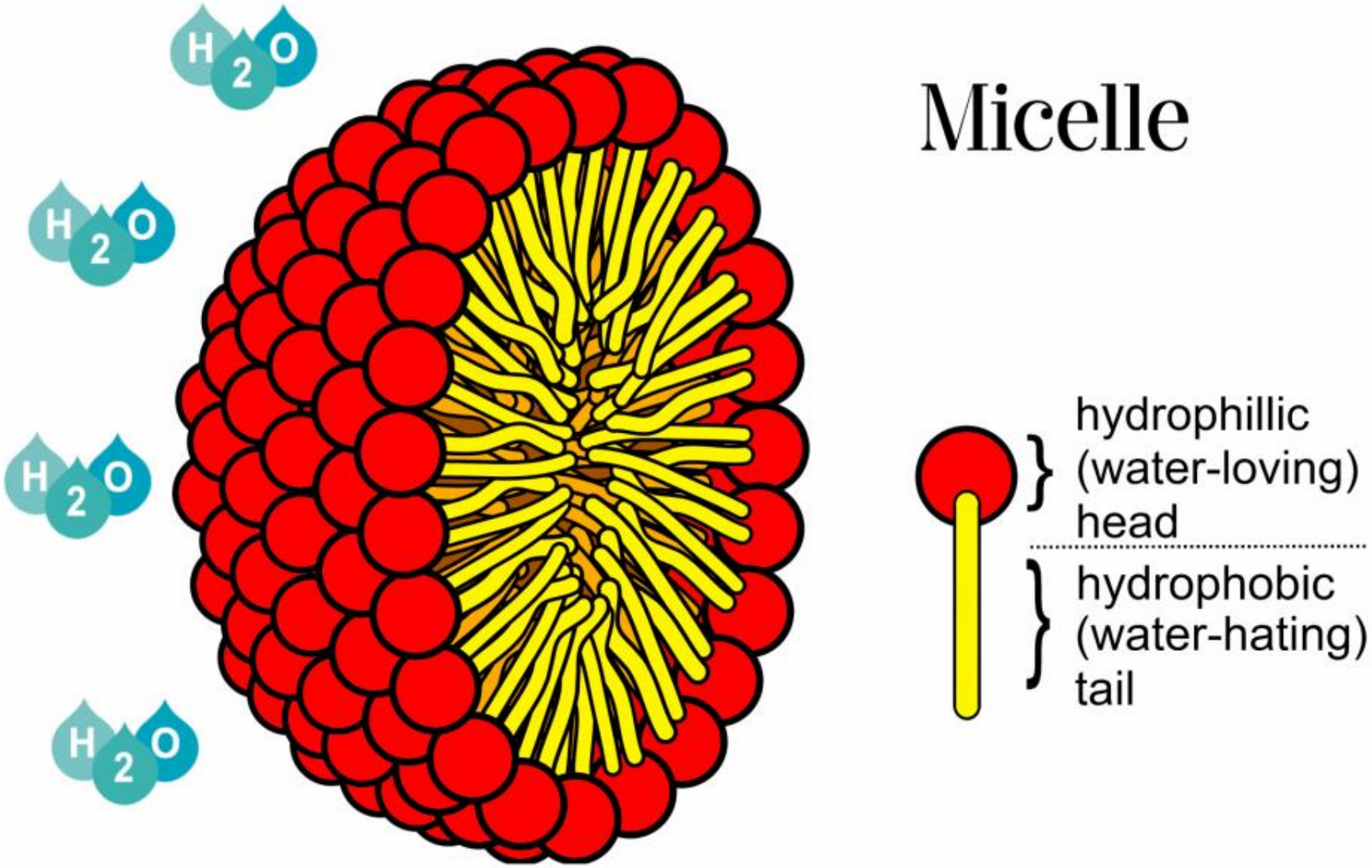


Work in a chemistry Lab: Miscible solvents

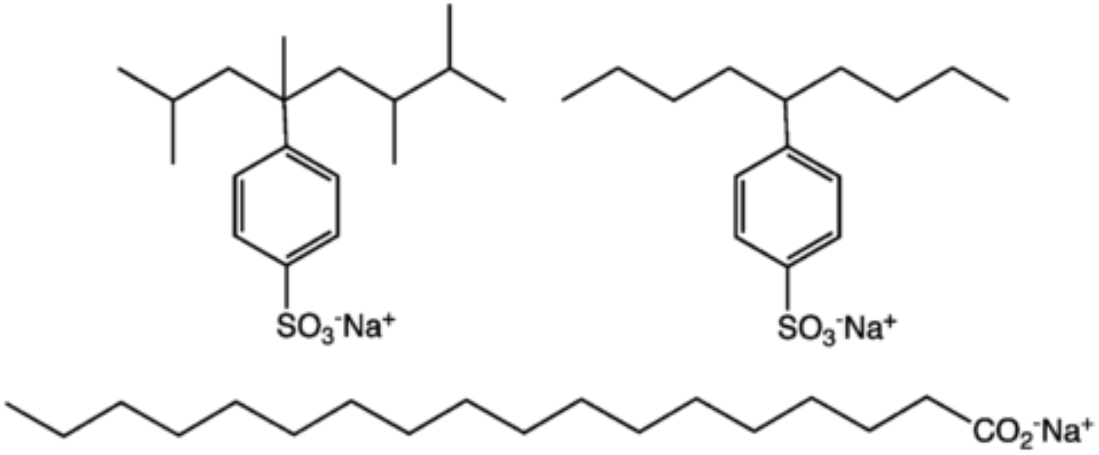
Solvent miscible table



Oil/ water mixture (heterogeneous mixture)



Laundry Detergent







Dr. Louis Ho Tan Tai



This book is available at FAS, USTH

Funny science…

A mixture of salt (NaCl), sand and I_2 >>>

a) Please suggest a strategy to separate components from the mixture?

b) Please suggest a strategy to purify I_2?

Si solar panel



Purification of Si from sand?

???

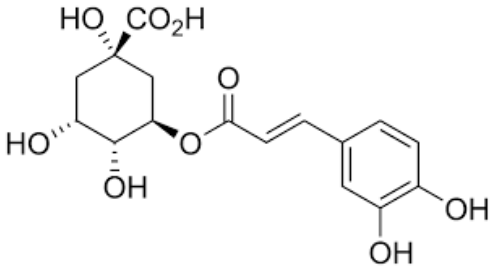
Phu Quoc, Vietnam



Complexity of a cup of coffee…

Chemically, coffee is remarkable complex and its exact composition depends on many of factors such as variety, roasting, grinding and brew conditions. The variability of coffee is underscored by a recent study which analyzed the chemical makeup of espresso coffees obtained from 20 different coffee shops in Glasgow, finding that the caffeine content ranged from 50 to 322 mg per cup. The most important chemicals in coffee are: chlorogenic acids, caffeine, diterpenes (cafestol and kahweol), trigonelline and melanoidins

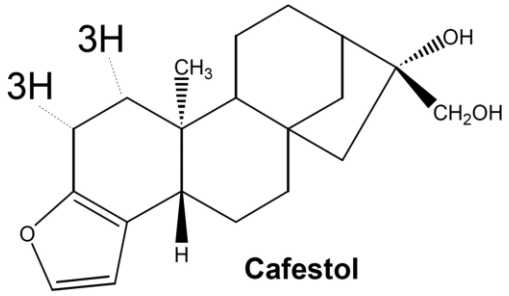




chlorogenic acid



caffeine



Cafestol

https://www.edwinvanbloois.com/2017/08/18/biochemistry-of-coffee/

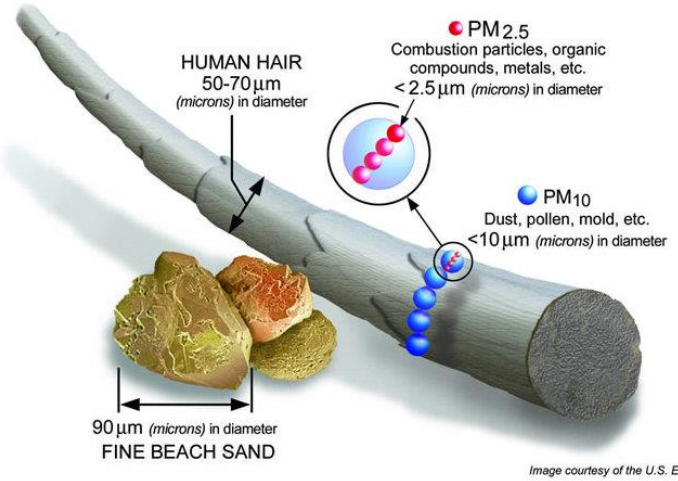
More complicated and critical issue: Air mixture

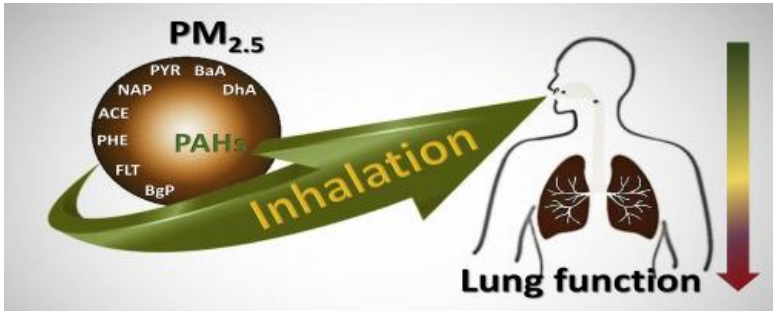
Hetero mixture



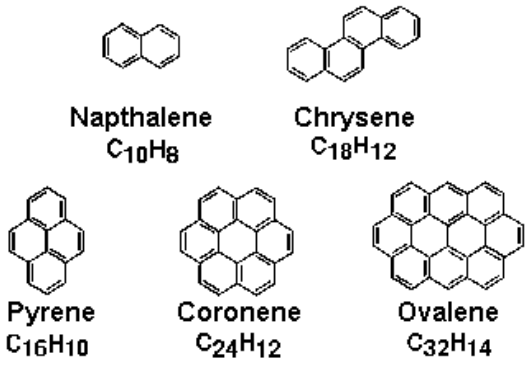


$PM_{2.5}$ particles





Polycyclic Aromatic Hydrocarbons



How could we remove the $PM_{2.5}$ particles from air?
(*Air purification*)

Air pollution with regard of high building…



(A)

Increase of
Particle amount?

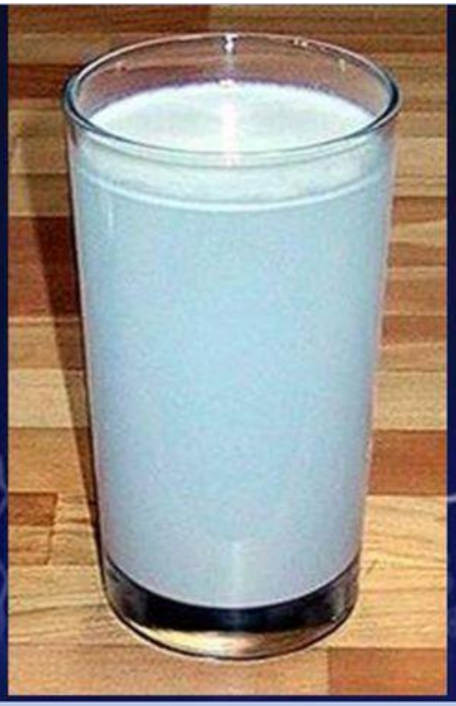
(B)

Decrease of
Particle amount?



Solution

Salt dissolved in water forms a uniform solution



Suspension

Flour suspended in water has a blue cast from light scattering off the flour particles

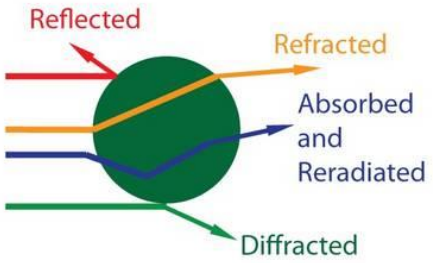


Colloid

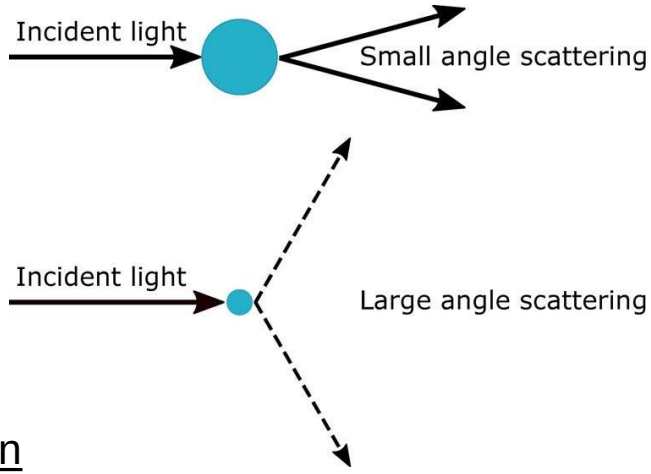
Milk contains butterfat globules dispersed in a watery liquid

Laser scattering particle size distribution analyzer

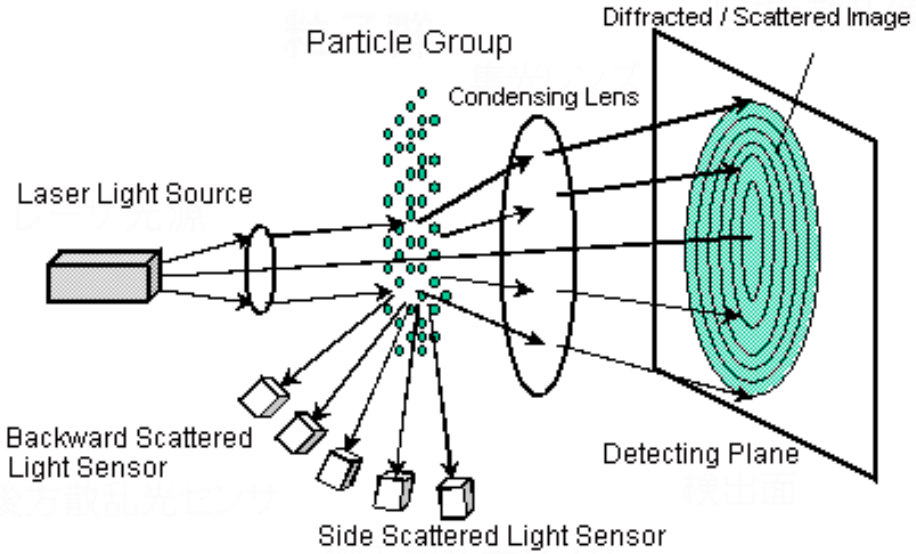
Laser – Particle interaction



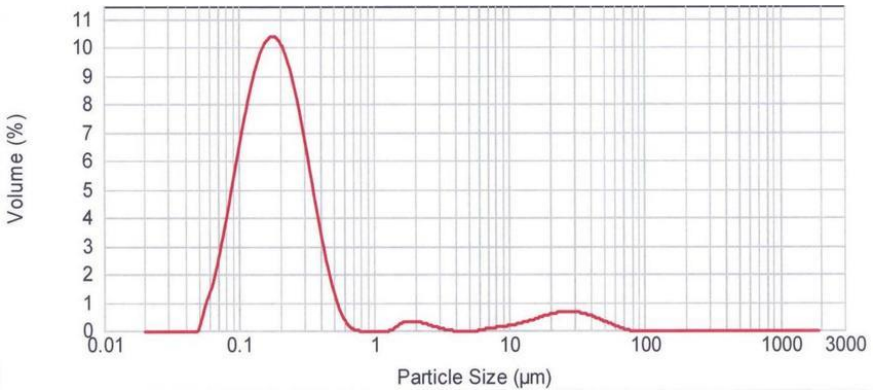
Interaction in function of particle size



Probing particle size distribution in suspension



A typical spectrum



https://www.mri.psu.edu/materials-characterization-lab/characterization-techniques/particle-sizing

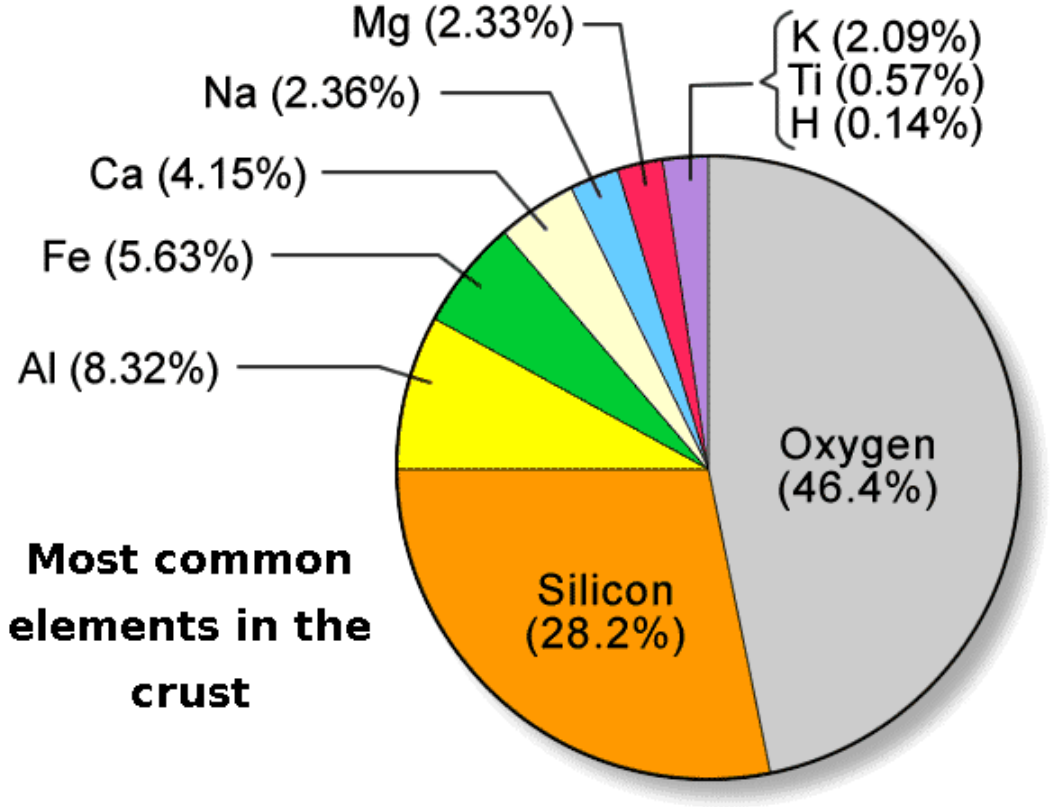
More discussion about mixture???

Element

(Smallest Substance, contains only one kind of atoms)

Can not be decomposed into simpler substance

Abundance elements in the Earth crust



Si solar cell



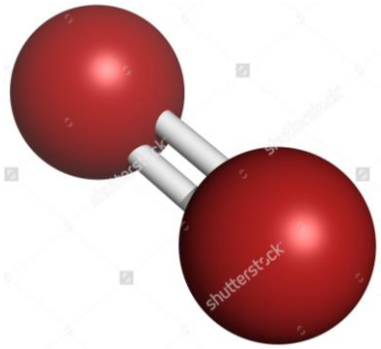
Compound

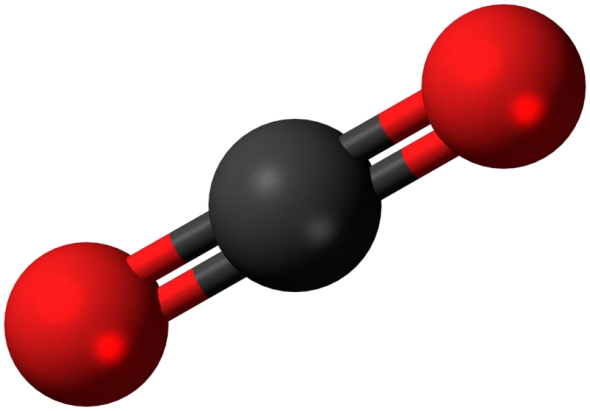
Substance composed of two or more elements

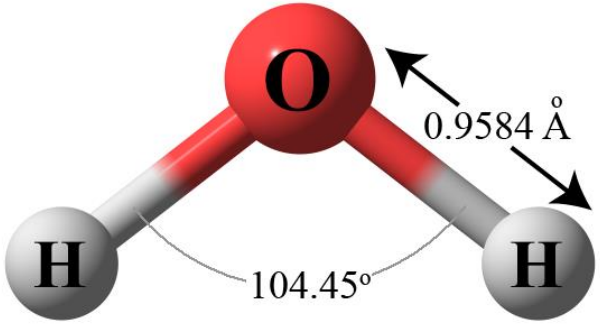
(Two or more kind of elements)

Can be decomposed by <u>chemical means</u> into simpler substances, always in the same ratio by mass

(H_2O through electrolysis form 2 H_2 and 1 O_2)





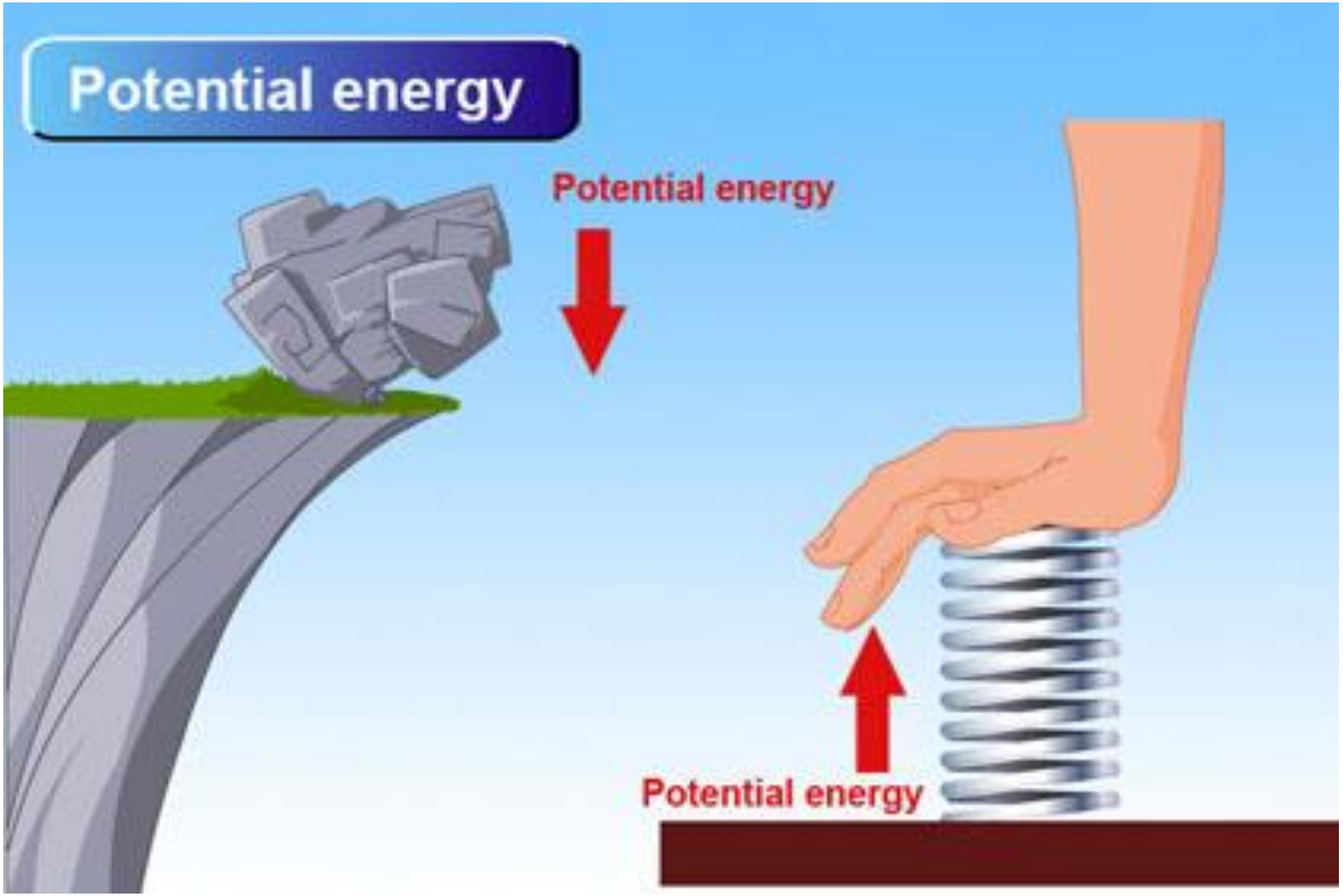


ENERGY

- **Energy** is defined as the capacity to do work or transfer heat
- Many **forms of energy**, including chemical energy,

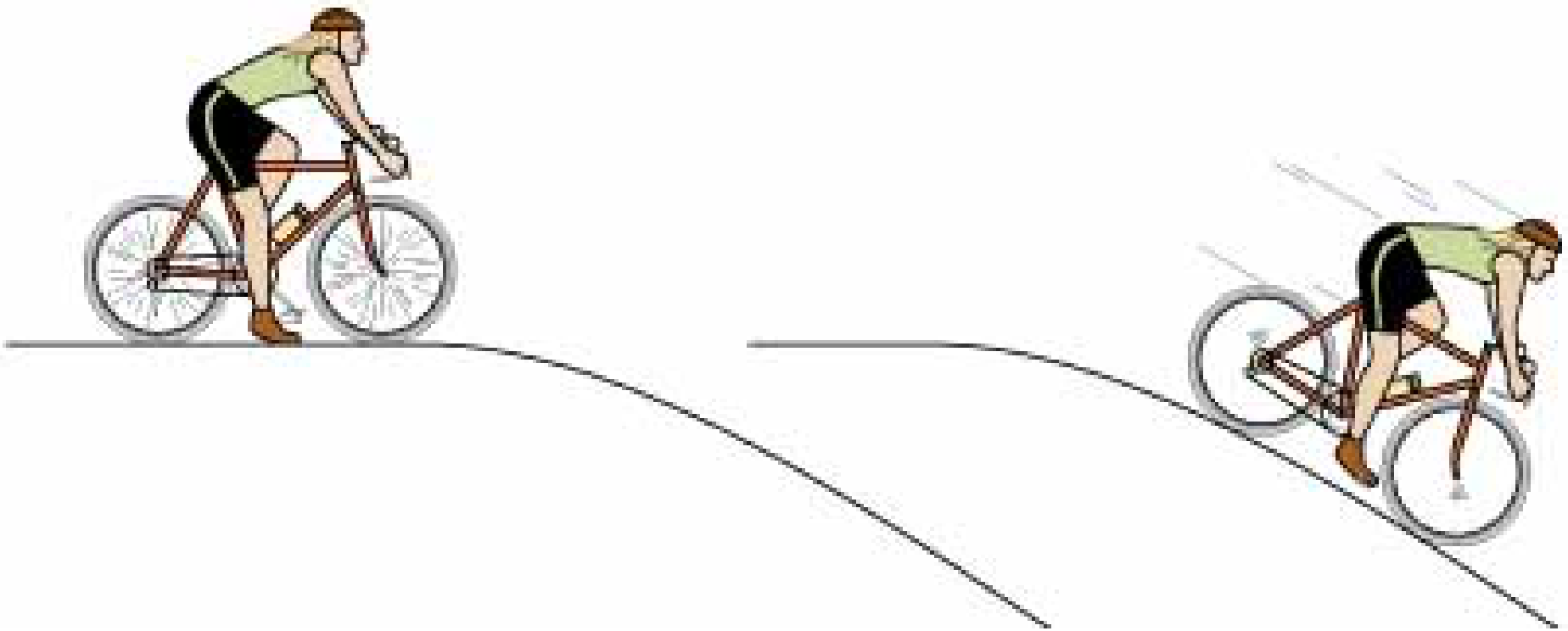
  light energy, electrical energy and heat energy
- Energy can be classified into 2 principle types:

  kinetic energy and **potential energy**



The energy that is stored in an object due to its position relative to zero position.

Kinetic energy



The energy associated with the motion of an object

ENERGY

POTENTIAL ENERGY

CHEMICAL ENERGY



GRAVITATIONAL ENERGY



NUCLEAR ENERGY



ELASTIC ENERGY



KINETIC ENERGY

MECHANICAL ENERGY



ELECTRICAL ENERGY



THERMAL ENERGY



LIGHT ENERGY



SOUND ENERGY

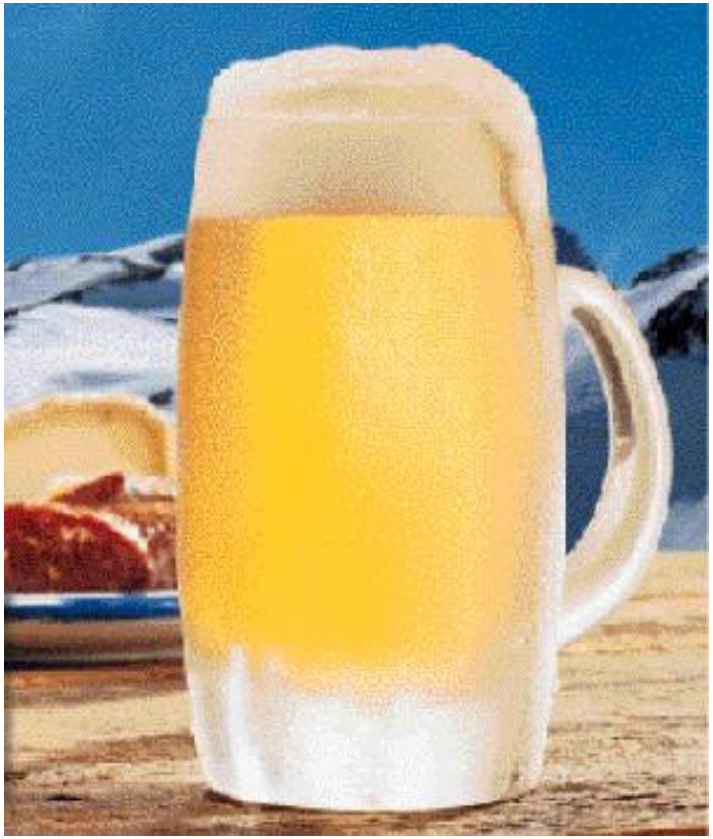


Exothermic and endothermic processes

Exothermic



Endothermic



Transportation of Energy from South to North: A dream?

Tet in Ho Chi Minh city



Tet in Ha Noi





“Gửi nắng cho em” by Pham Tuyen

…wishes to send “solar energy” from HCM city to Hanoi…

How does chemistry help to realize this dream?



Transportation of Energy from Summer to Winter: A dream?

Very hot summer



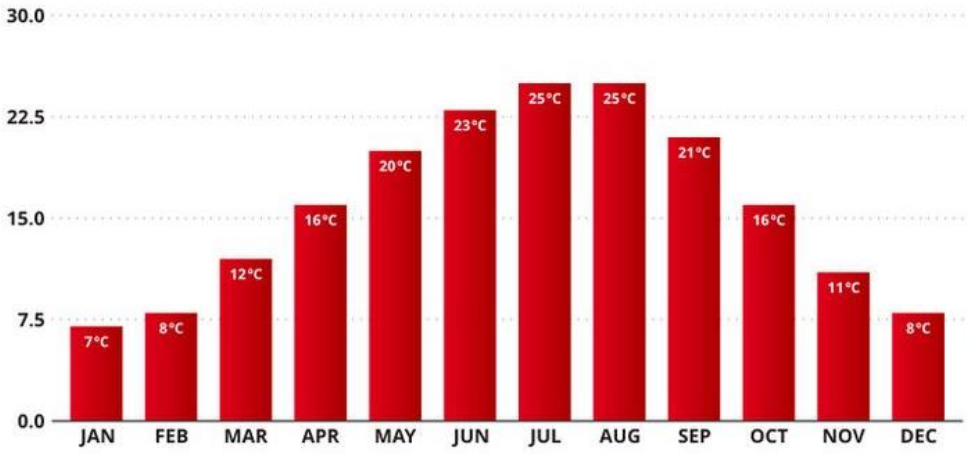
Heat transfer
Season to season

Energy storage

(Extreme) Cold winter



Temperature evaluation over a year in Paris



Thermochemical energy storage: Based on a “simple” reaction

$$A_{(s)} + B_{(g)} \leftrightarrow AB_{(s)} + \Delta H$$

@ By tuning the B vapor $\rightarrow$ Tune the Heat formation

Requirements:

@ High ΔH value

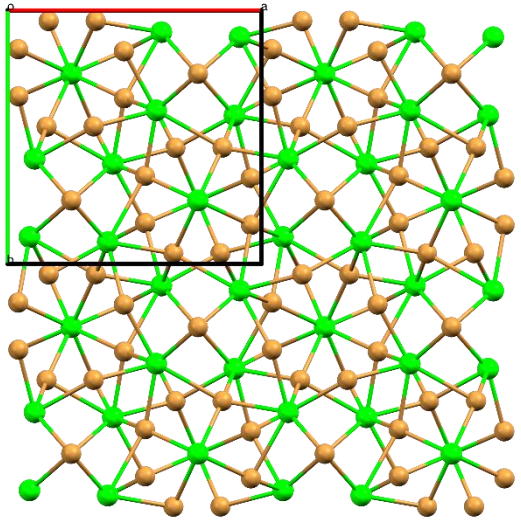
@ Reversible reaction

@ Fast kinetics

@ Cheap, made of abundant elements

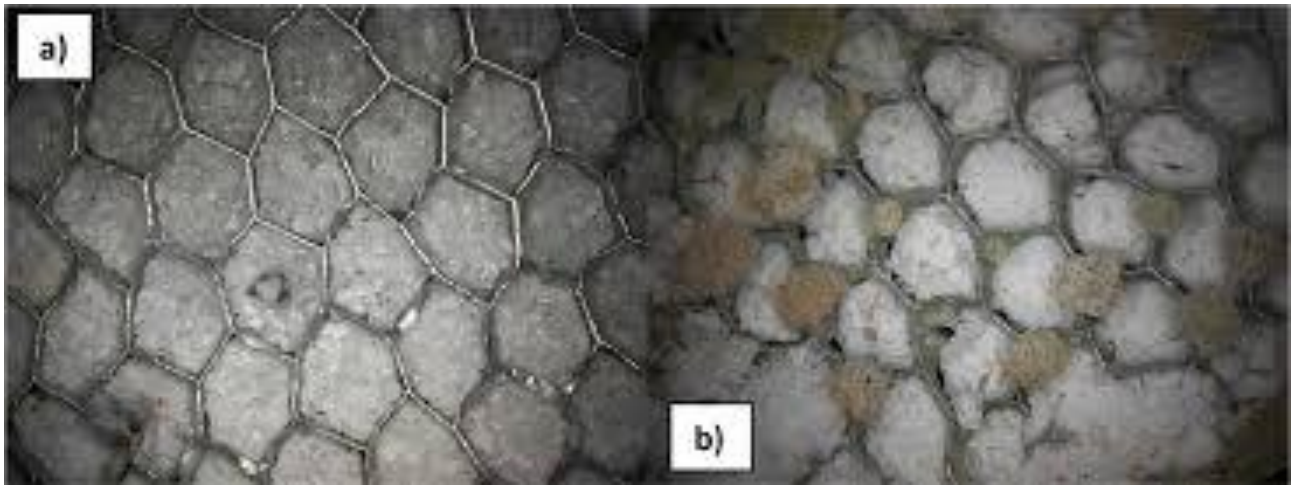
An example: **$SrBr_2$** salt

$SrBr_2$ crystal structure



$SrBr_2 + 6H_2O \leftrightarrow SrBr_2(H_2O)_6 \quad \Delta H$ 3744 KJ/Kg H_2O

Ideal energy storage density is very high: 628 kW h m^{-3}

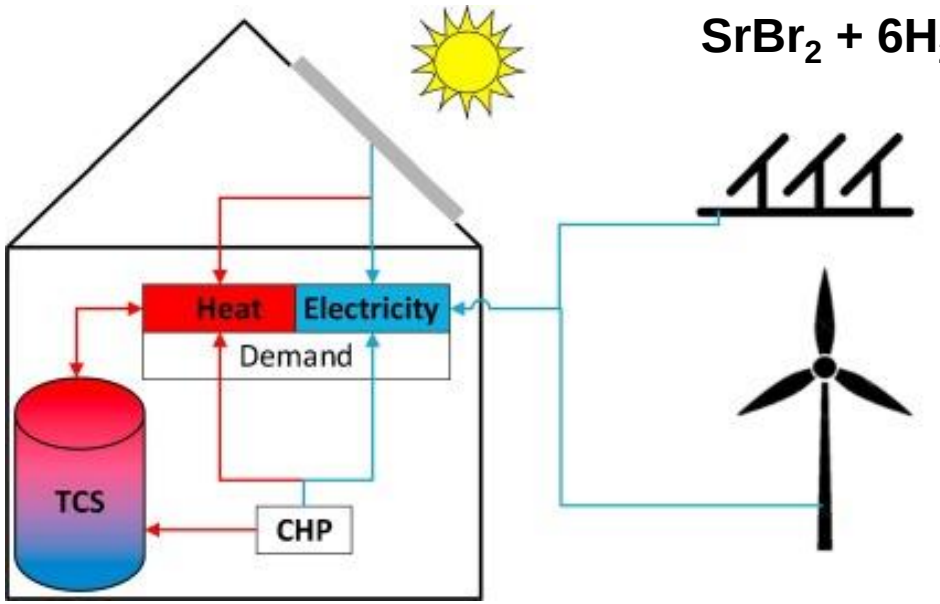


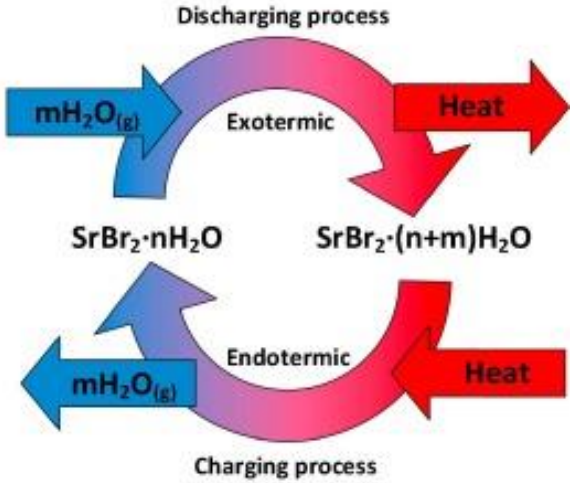
$SrBr_2$ in the (a) hydrate form and the (b) dehydrated form in the honeycomb structure heat exchanger

An example: $SrBr_2$ salt

(Winter: Collect heat)

$$SrBr_2 + 6H_2O \rightarrow SrBr_2(H_2O)_6 \quad +3744\ KJ/Kg\ H_2O$$





$$SrBr_2 + 6H_2O \leftarrow SrBr_2(H_2O)_6 \quad +3744\ KJ/Kg\ H_2O$$

(Summer: Remove heat)

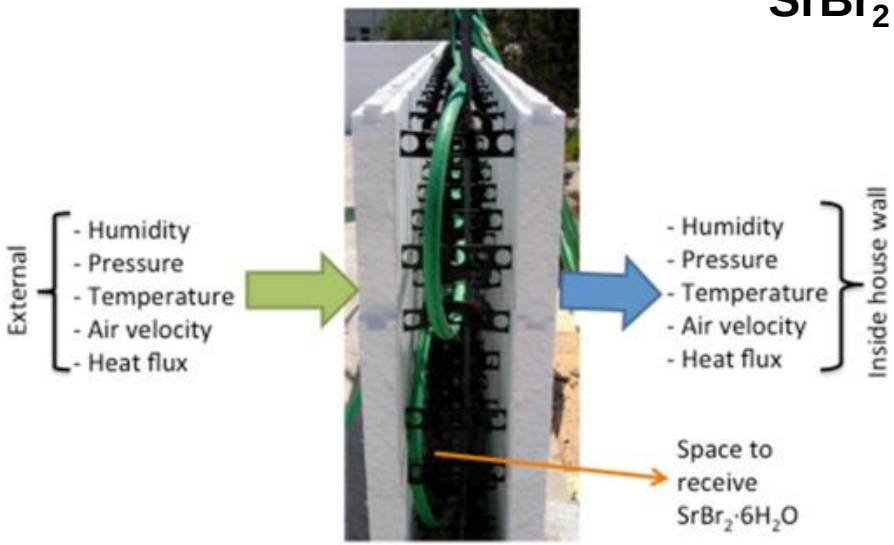
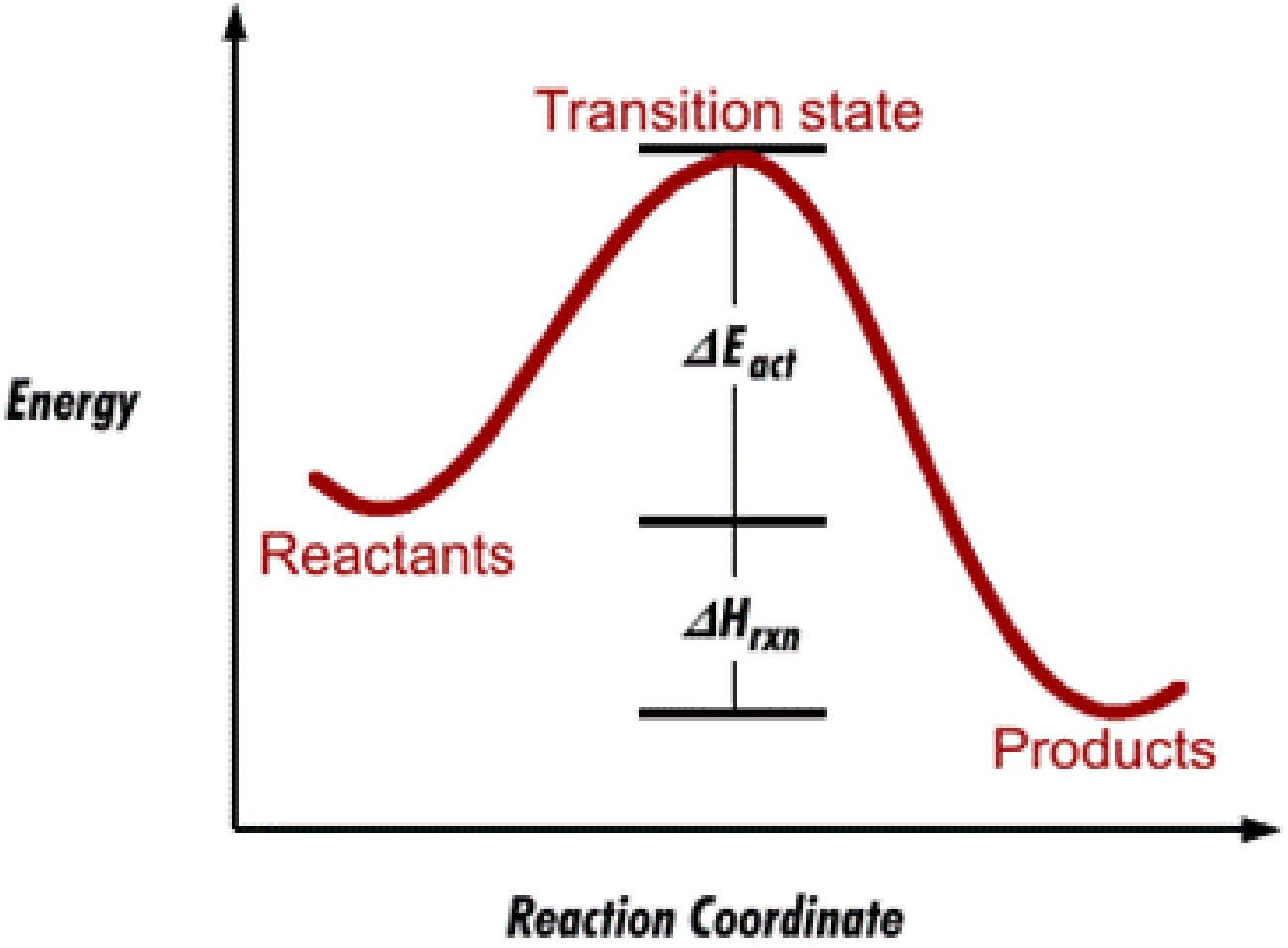


Illustration of building wall integrated storage material for thermal comfort

Think about Chemical Reaction: Think about Energy

(*Cost to do so*)



Details: in General Chemistry II

- **<u>The law of conservation of matter</u>**: there is no observable change in the quantity of matter during a chemical reaction or during a physical change
- **<u>The law of conservation of energy</u>**: energy cannot be created or destroyed. It can only be converted from one form to another
- The **<u>combined amount</u>** of matter and energy in the universe is fixed
- The relationship between energy and matter:

$$\mathbf{E = mc^2}$$ (c is the speed of light)

- **Chemical properties** are exhibited by matter as it undergoes **changes in composition**

  (Valence electrons NOT nuclei)

- All substances also exhibit **physical properties** that can be observed in the absence of any change in composition
  - Color, density, hardness, melting point, boiling point and electrical and thermal conductivities

- **In any physical change** (3 states of matter):

  no change in chemical composition

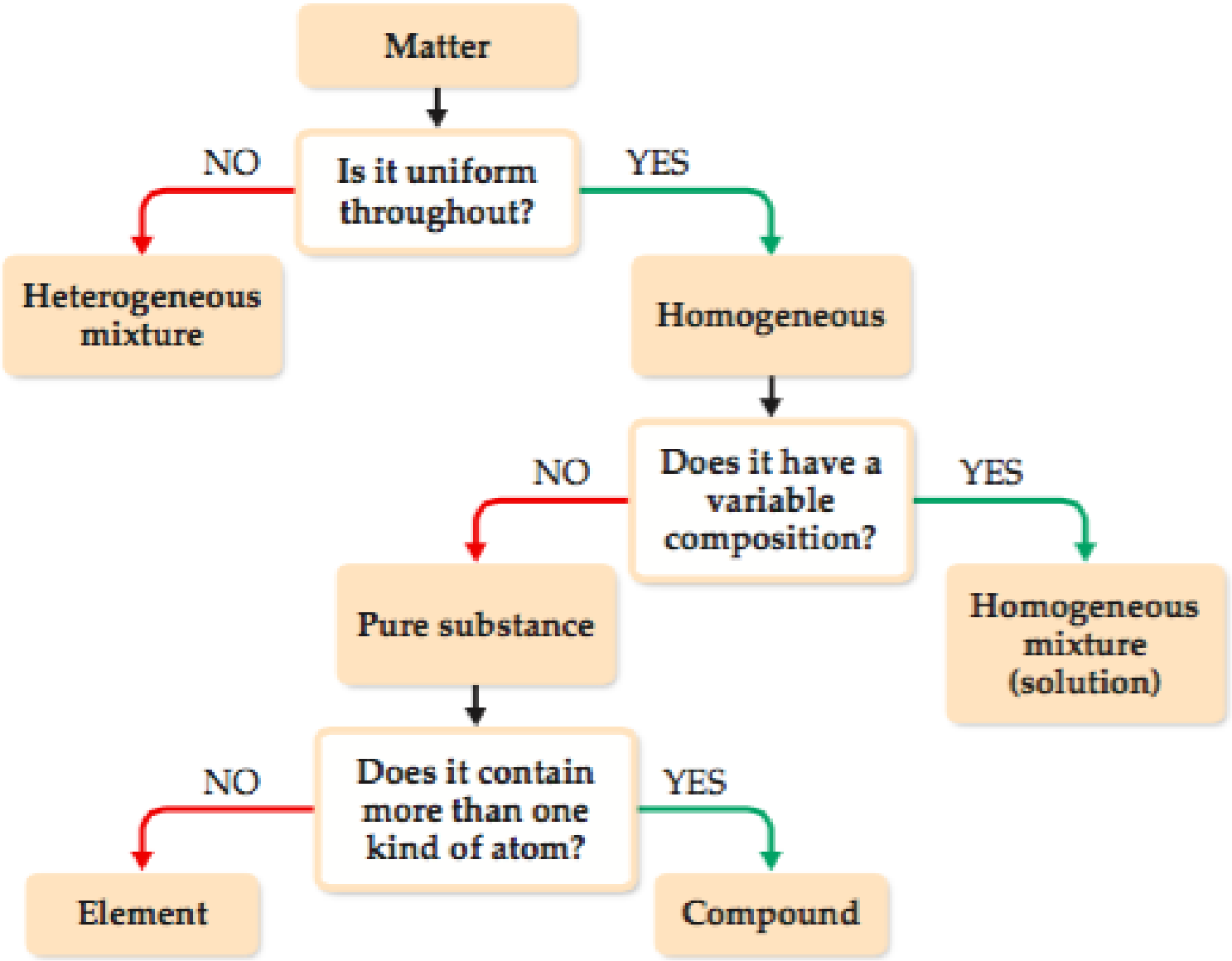
- **In any chemical change**
  - One or more substances are used up
  - One or more new substances are formed
  - Energy is absorbed or released

What will happen if the energy difference is Zero?

($\Delta G^0 = 0$ KJ/mol)

What is chemical reaction having $\Delta G^0 = 0$ KJ/mol?

Classification of Matter



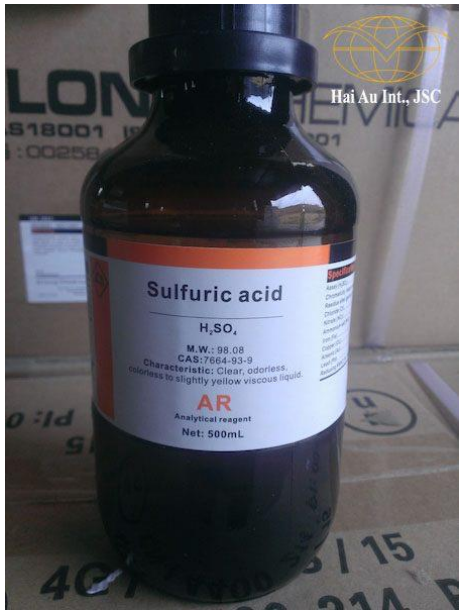
Simple examples…

Sand + Salt + Water



Hanoi beer





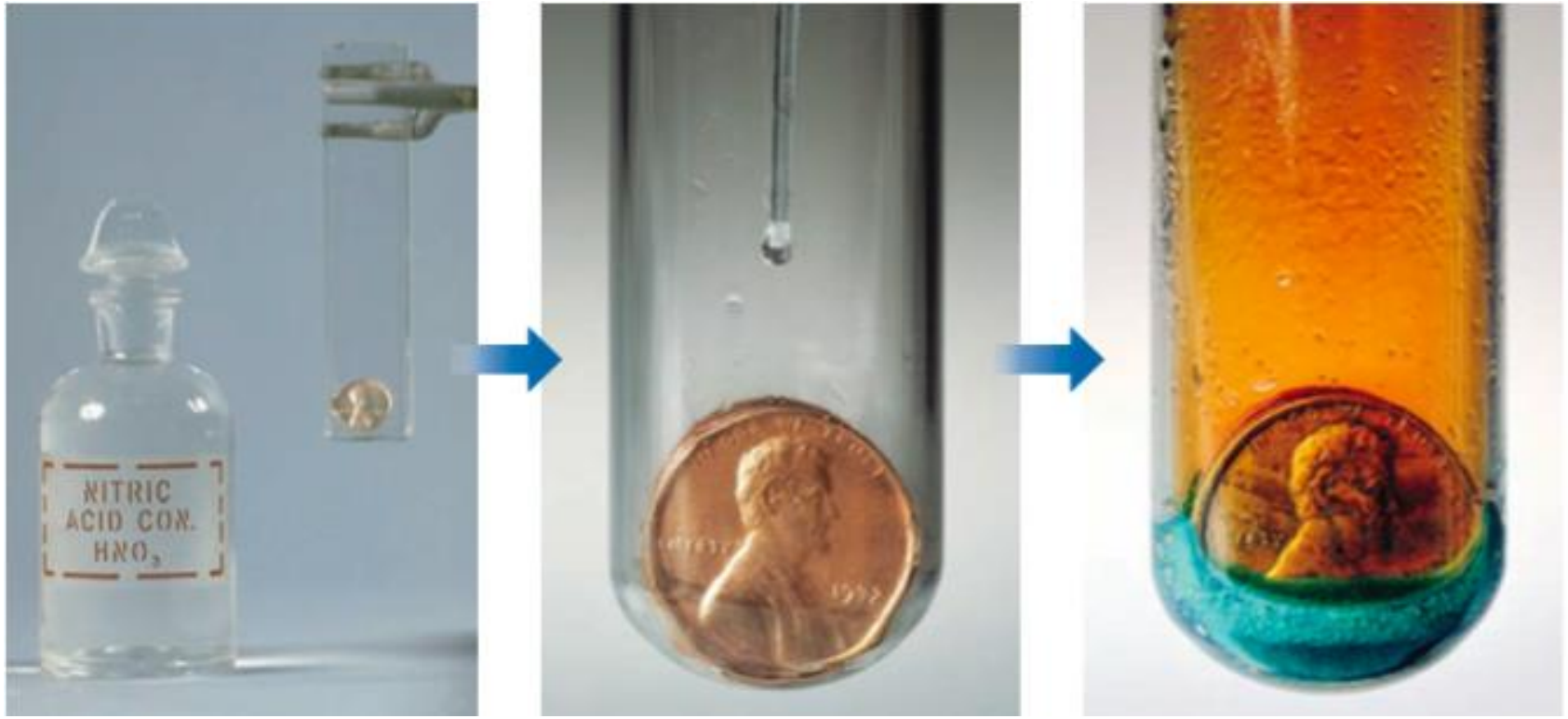
Sulfur powder



Chemical change (Chemical reaction)

A substance is transformed into a chemically different substance

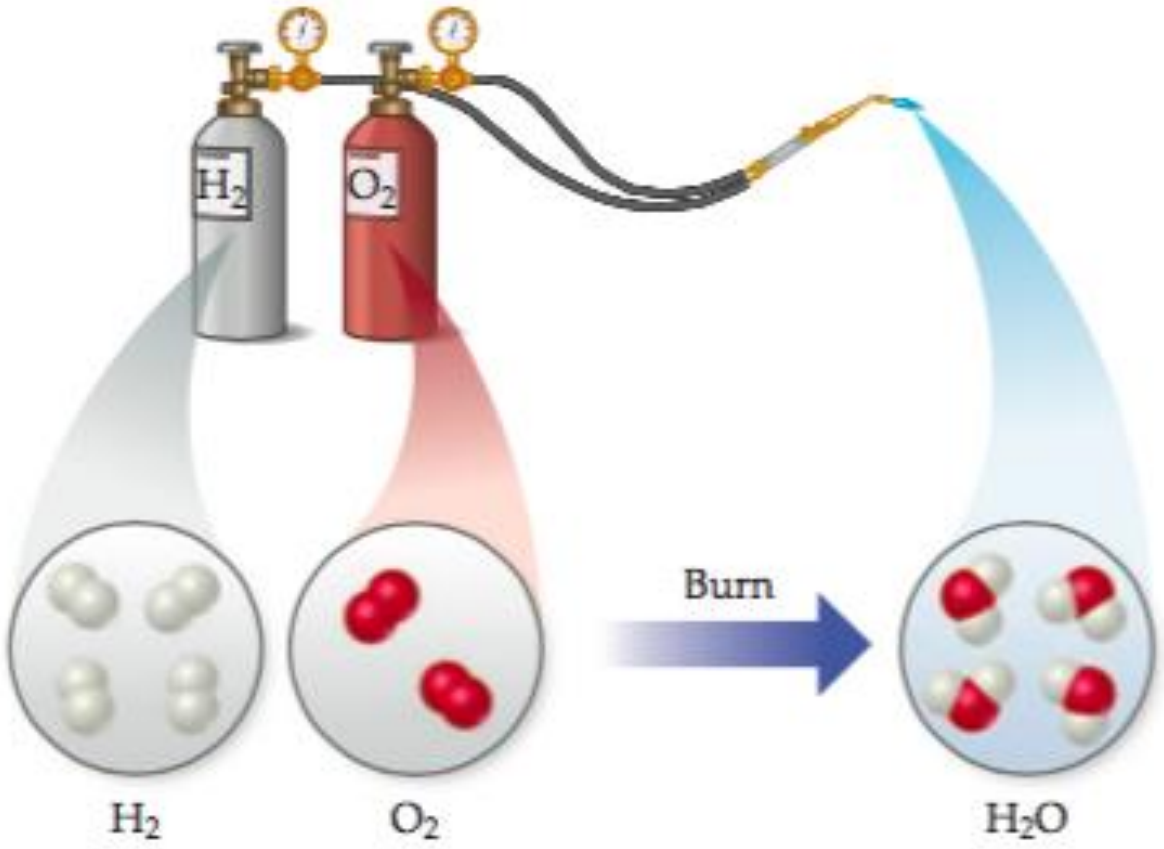
A ➔ B



Chemical change (Chemical reaction)

A substance is transformed into a chemically different substance

A ➔ B

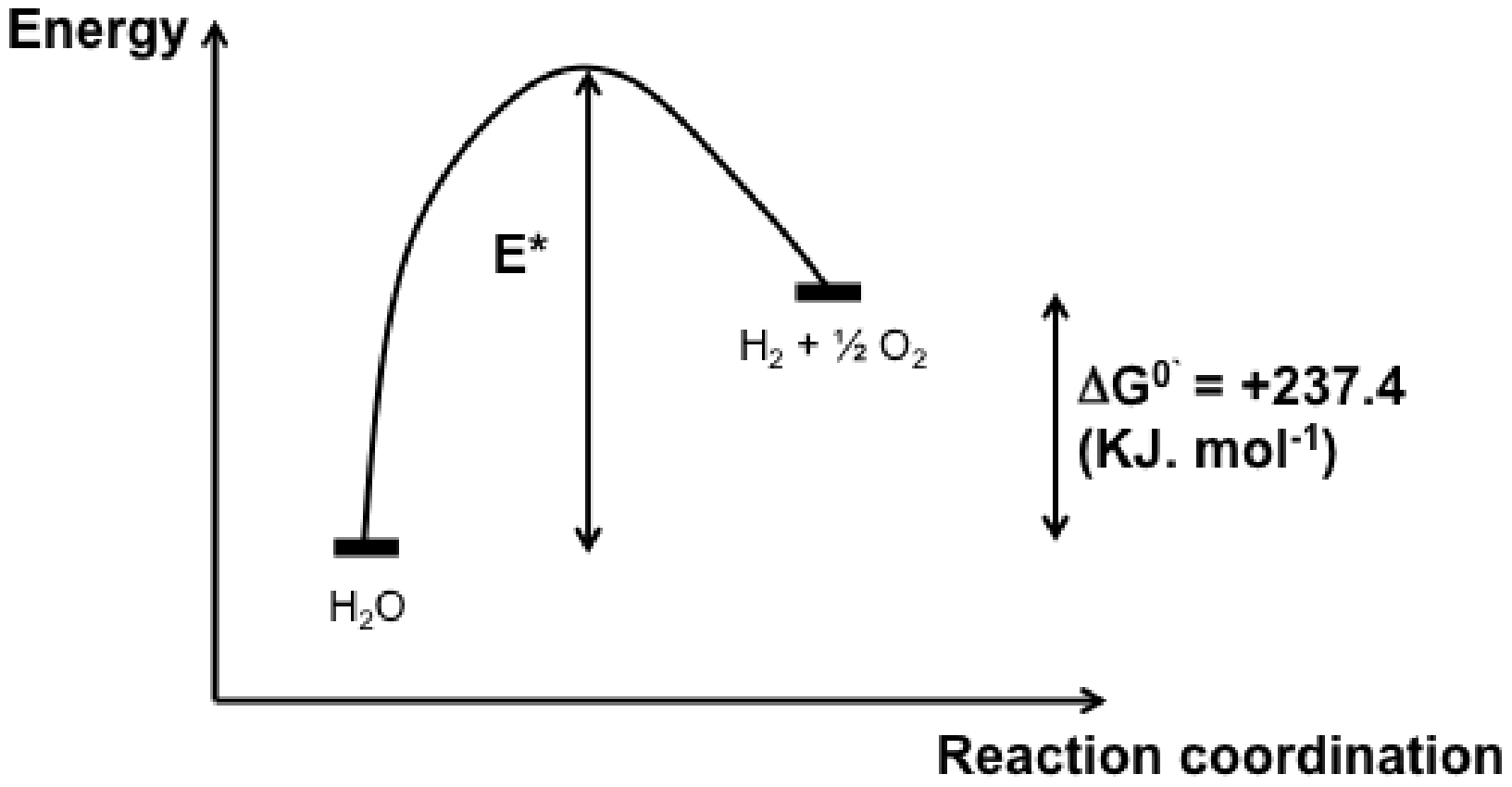


Exothermic reaction

ΔG^0 = -273 KJ/mol

Water formation is a **down hill reaction**.

Can we make a mixture of H_2 and O_2 (without chemical reaction ?)



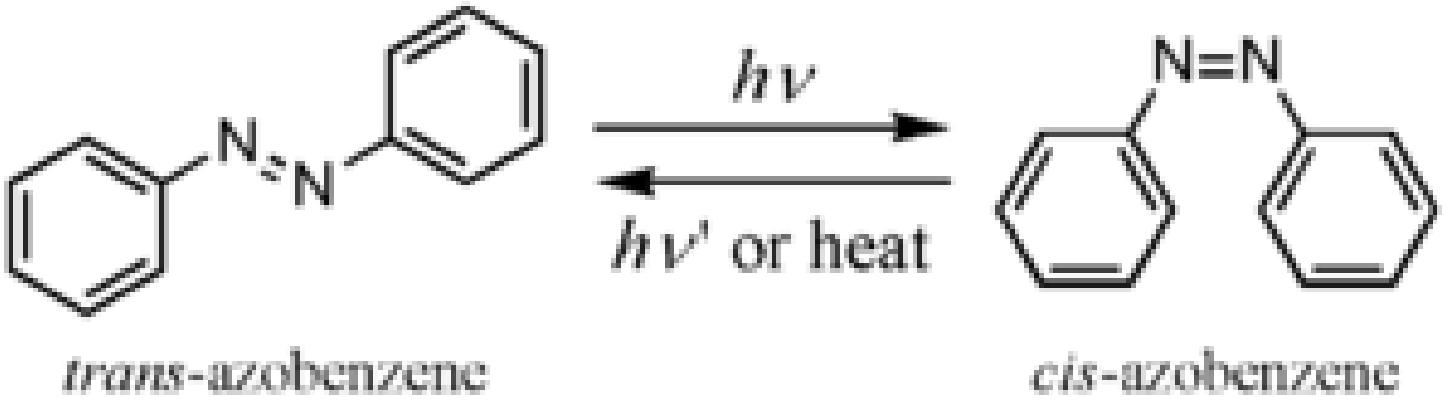
Slow kinetic reaction

Chemical change (Chemical reaction)

A substance is transformed into a chemically different substance

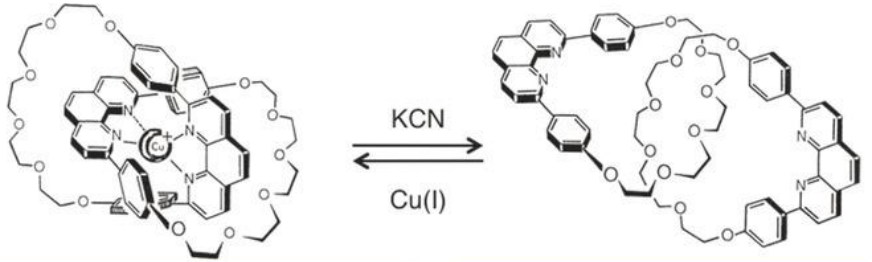
A ➔ B

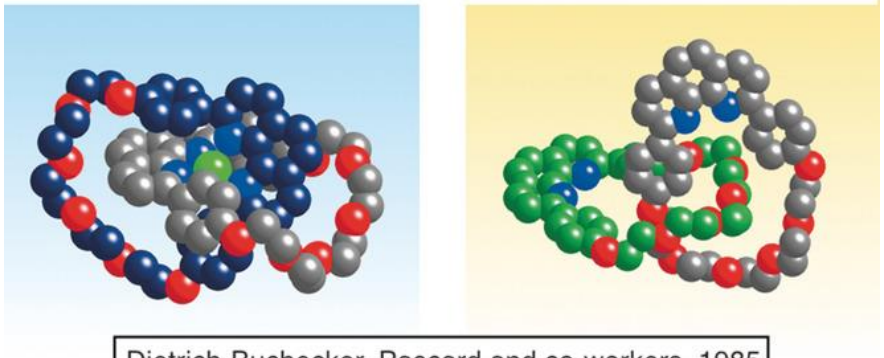
(even B and A have identical chemical composition)



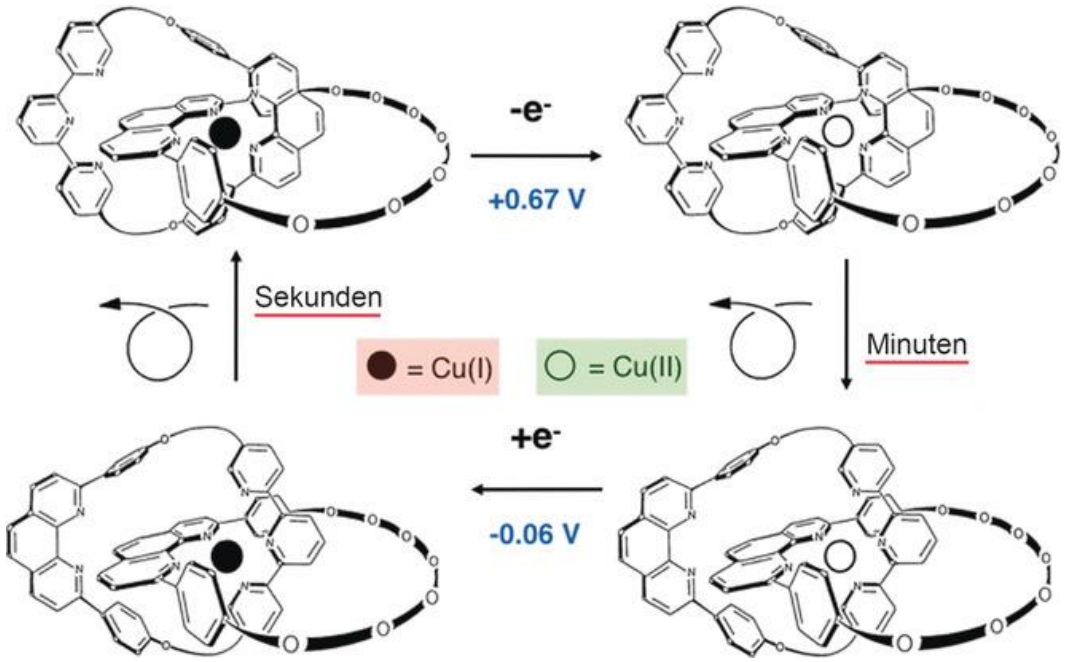
Molecular machine (Nobel prize 2016, Chemistry): *Artificial muscle*

Molecular machine (Nobel prize 2016, Chemistry)





Dietrich-Buchecker, Pascard and co-workers, 1985



Sulfur powder is:

a) A heterogeneous mixture

b) A homogeneous mixture

c) A compound

d) An element

Which of these changes are physical and which are chemical? Explain.

A. Plants make sugar from carbon dioxide and water.

B. Water vapor in the air forms frost.

C. A goldsmith melts a nugget of gold and pulls it into a wire.

D. Nickel nanoparticles (diameter of 10 nm) burn in air

Hydrogen is considered to be a promising energy carrier (a clean fuel). It is in the gas phase at the ambient conditions (Room temperature and atmospheric pressure).

To liquidify H_2 (transform it into liquid), one can:

a) Reduce pressure and increase temperature

b) Increase pressure

c) Increase pressure and reduce temperature

d) Reduce pressure and reduce temperature

Burning gasoline to power a car is a:

a) Chemical – to – motion energy conversion process

b) Motion – to – chemical energy conversion process

c) Efficient energy conversion with the conversion yield of 100%

d) Endothermic process

e) Exothermic process

Describe the separation method(s) involved in brewing a cup of coffee.



With sufficient energy input (*e.g.* thermal or solar energy), water is evaporated but not decomposed into H_2 and O_2. Give your explanation.

A USTH student aims to conduct a synthetic experiment at temperature of -10 ^{o}C. How can he/ she do?

a) Use an ice bath

b) Use an ice/NaCl bath

c) Use a liquid nitrogen bath

d) Use dry-ice bath

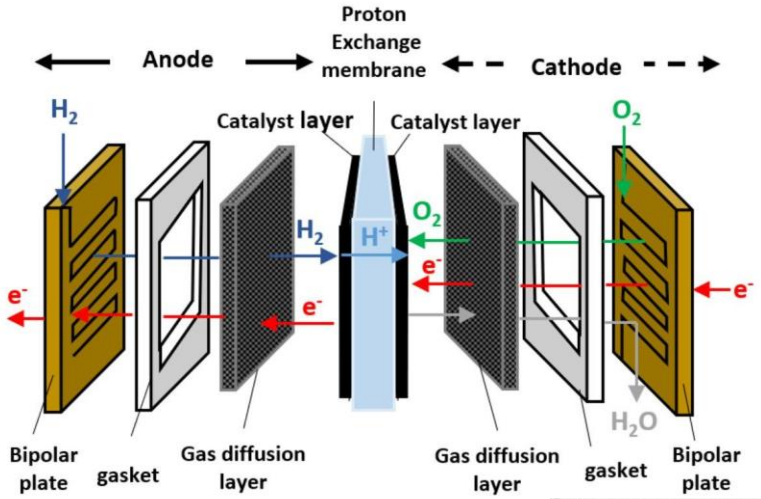
Mixture H_2, CO: Purification Challenge

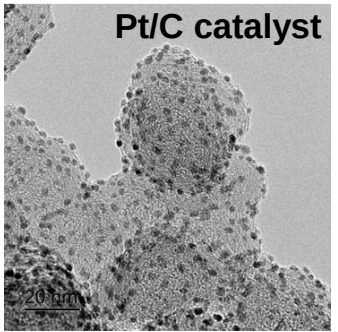
H_2 produced by reforming natural gas contains CO impurity which is harmful to Pt catalyst in the proton exchange membrane fuel cell (*e.g.* Toyota H_2 car).

Please suggest a strategy to eliminate CO impurity from H_2 gas.



Proton exchange membrane fuel cell





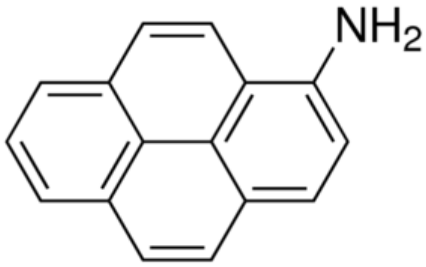
Tutorials (additional)

Suggest a procedure to separate following compounds from a mixture

Graphene sheets

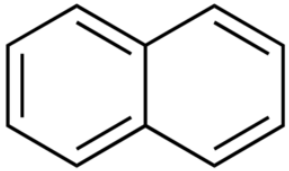
1-Pyrenemethylamine

Naphthalene



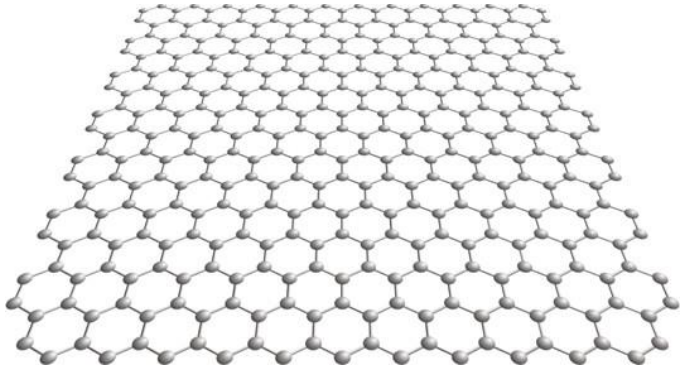
1-Pyrenemethylamine

+



Naphthalene

+



Graphene sheets

Exhausted gas from car



N_2
O_2
H_2O
CO_2
CO
NO_x (Nitrogen oxides)
SO_2
Hydrocarbons, Benzene
Black carbon (powder)

1. Is exhausted gas homogeneous or heterogeneous?

2. How can we *clean up* the exhausted gas (to make it less toxic)?

3. How can we *REDUCED* the exhausted gas?

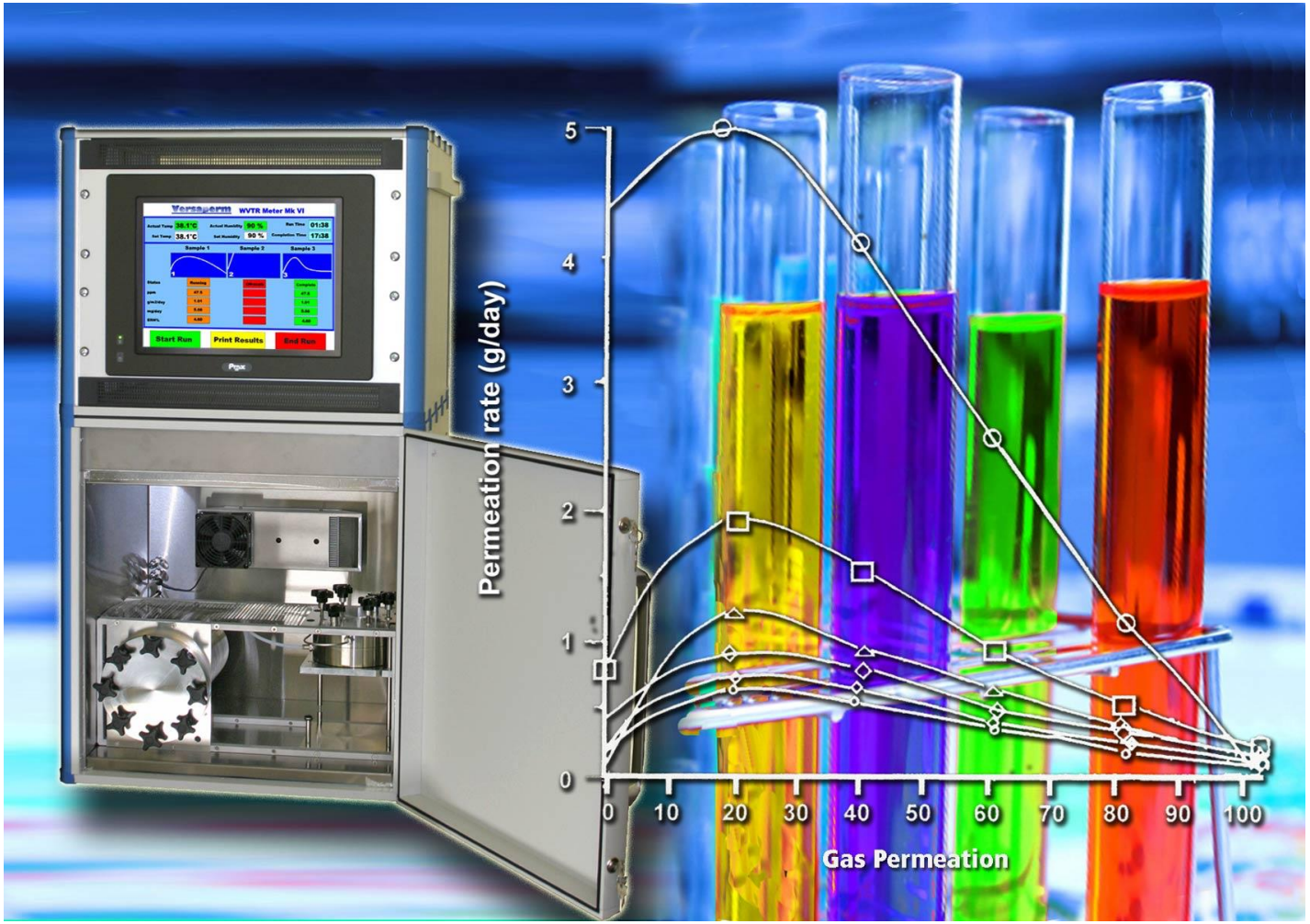
In Vietnam, groundwater is found to be contaminated by As, Fe, Mn, and NH_4^+ pollutants. The arsenic levels ranged from 0.1–3050 µg/L, which substantially exceed the standard of 10 µg/L established by the WHO.

The primary form of arsenic found in the groundwater is As(III) with some As(V).

$$H_3AsO_3 + H_2O \leftrightarrow H_2AsO_4^- + 3H^+ + 2e^-$$
$$H_3AsO_3 + H_2O \leftrightarrow HAsO_4^- + 4H^+ + 2e^-$$

1. Please suggest a potential strategy to eliminate As contaminant?

Work in Laboratory: Measurement, Measurement, …



Units of measurement

SI Units: adopted in 1960 based on an international agreement

(SI units = Systeme International d`Unites)

SI base units

Base quantity	Name	Symbol
length	meter	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
thermodynamic temp	kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd

Prefixes used in the Metric System and with SI units

Prefix	Symbol	Decimal Equivalent	Power of 10
mega-	M	1,000,000	Base x 10^{6}
kilo-	k	1,000	Base x 10^{3}
deci-	d	0.1	Base x 10^{-1}
centi-	c	0.01	Base x 10^{-2}
milli-	m	0.001	Base x 10^{-3}
micro-	μ or mc	0.000 001	Base x 10^{-6}
nano-	n	0.000 000 001	Base x 10^{-9}
pico	p	0.000 000 000 001	Base x 10^{-12}

Temperature: a measure of the hotness or coldness of an object

Celsius scale

(Assignment of 0°C to the freezing point of water and 100°C for its boiling point)

Kelvin scale (SI)

(0K refers to the lowest attainable temperature, absolute zero)

$$\mathbf{K = °C + 273.15}$$

Fahrenheit scale (USA)

(Water freezes at 32 °F and boils at 212 °F)

$$\mathbf{°C = 5/9\ (°F - 32)}$$

Derived SI Units

A derived unit is obtained by multiplication or division of one or more of the base units

Volume

Density

Charge

etc

Uncertainty in Measurement

https://www.youtube.com/watch?v=d-Fv682oMZg

Exact numbers: those whose values are know exactly

(*e.g.* a dozen of eggs = 12 eggs; number of B1 students)

Inexact number: those whose values have some incertainty

Note: All numbers from experiment are inexact

+ Equipment errors

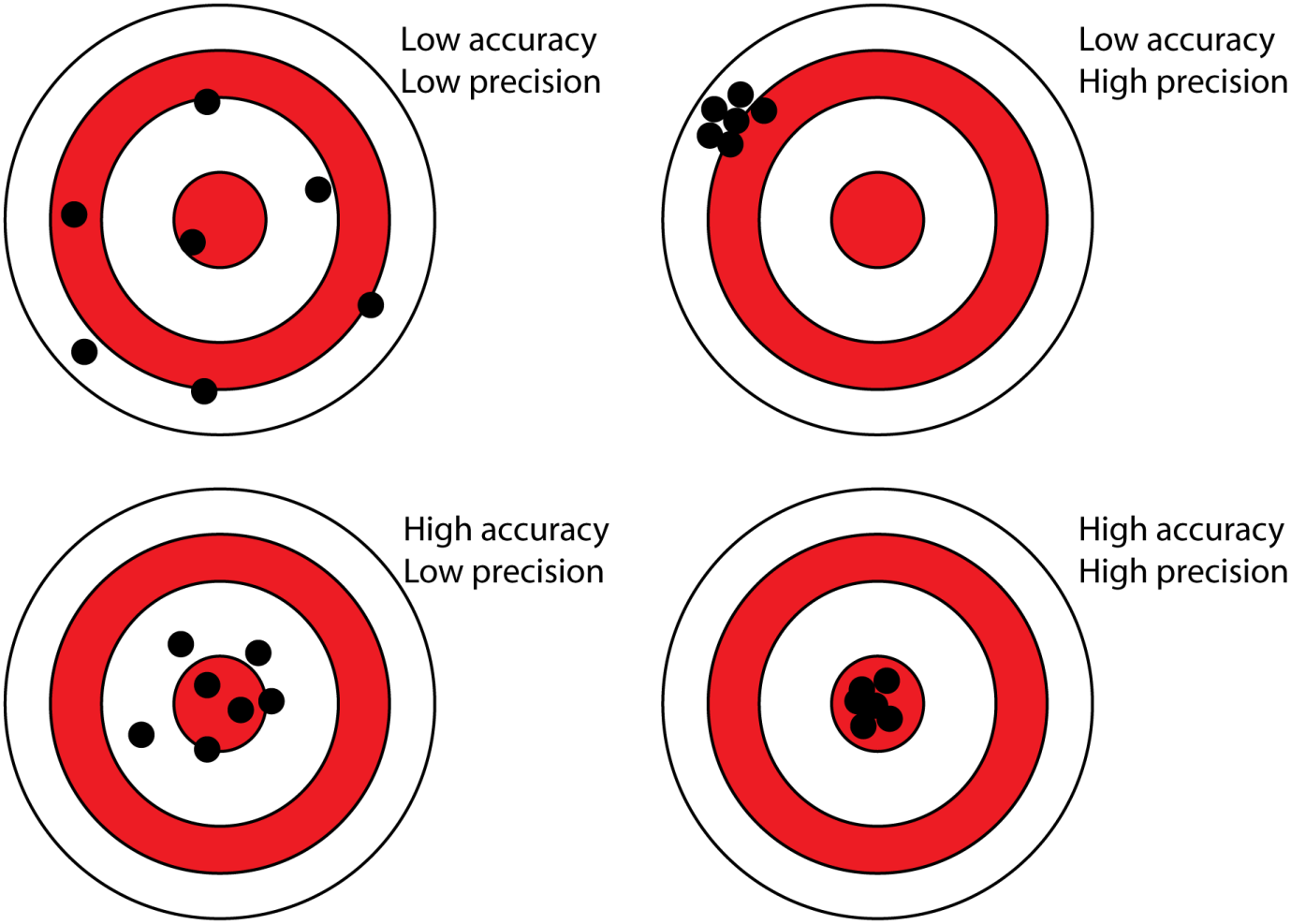
+ Human errors

Precision:

a measure of how closely individual measurements agree with another

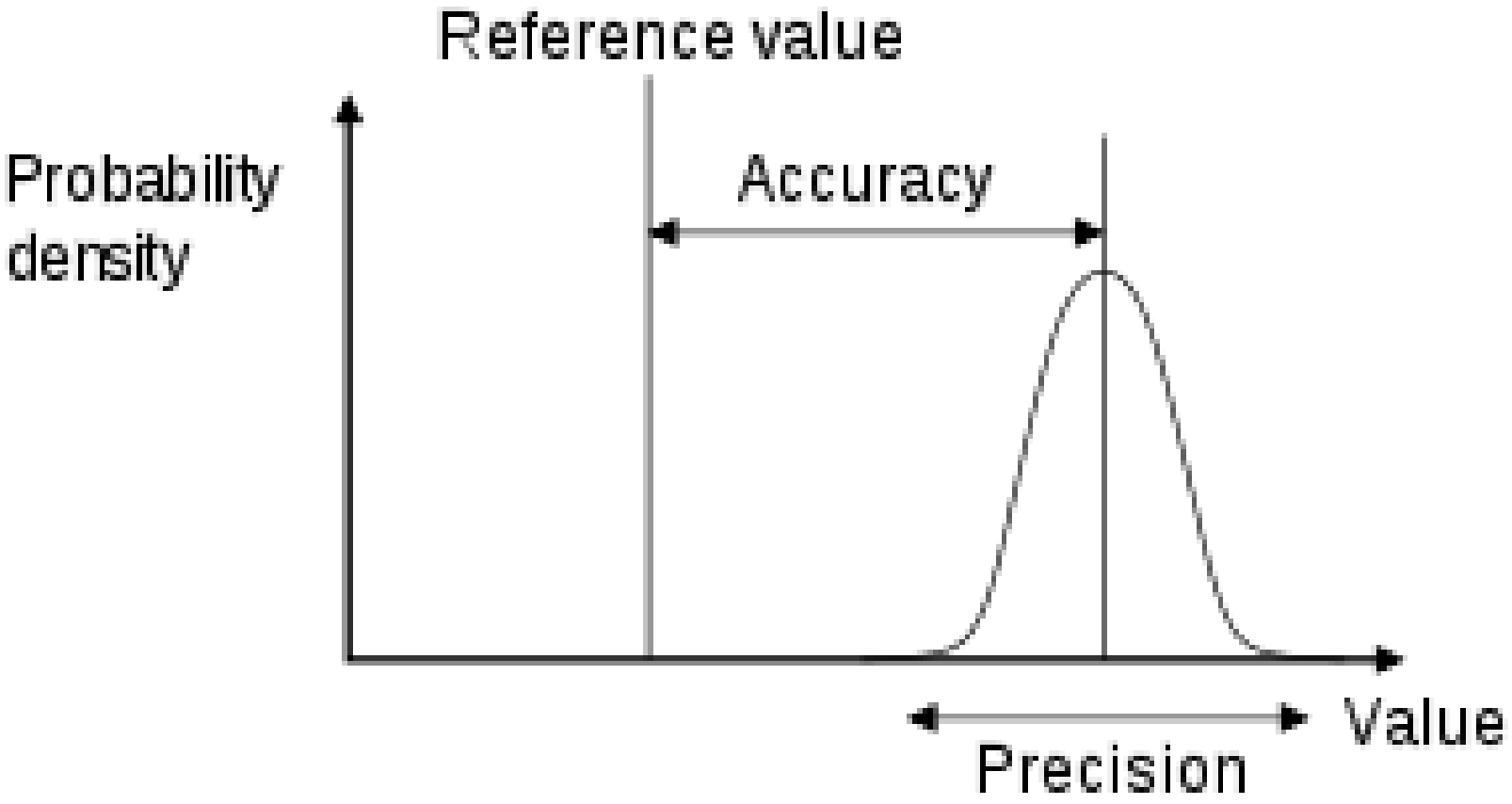
Accuracy:

How closely individual measurements agree with the correct, or “true” value



To have a good accuracy: Calibration is demanded

Thus, it is critical to put/ use a correct reference



But, for a new measurement where the reference value is not available (or not yet widely accepted)

What should we expect? Accuracy or Precision?

Analytical balance



There is always some uncertainty in the last digit reported for any measured quantity

2.2405 ± 0.0001

500 brothers:

Meaning: It is neither 400 nor 600

But it can be from 501 to 599

Thus, it is important to report about uncertainty of measurement

All digits of a measured quantity, including the uncertain ones, are called **significant figures**

4.0	2 significant figures
4.05	3 significant figures

4.0± 0.1

4.05 ± 0.01

Rules:

* Zeros (between non zero digits): Significant

* Zeros (at the beginning of a number): Never Significant

They indicate the position of the decimal point

0.002 Only one Significant

* Zeros (end of a number): Significant if the number contains a decimal point

Significant figures in calculation

Rule for addition and subtraction

* The result has the same number of **decimal places** as the measurement with the fewest decimal places

* When the result contains more than the correct number of significant figures ➔ it must be rounded off.

$$1.322 + 20.42 + 83.\mathbf{1} = 104.\mathbf{8}\underline{42}$$

Thus, the result to be reported: 104.8

Rule for multiplication and division

The result contains the same number of significant figures as the measurement with the fewest significant figures

6.221 x 5.2 = 34.3492

4 significant figures

2 significant figures

Should be rounded off to 2 significant figures

Thus, result to be reported: **34**

But NOT **34.0**

When round off the number:

34.3492 ➔ 34

34.6174 ➔ 35

General Chemistry I

Chemistry: The central science

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